

# The Visible Fourier Ring Is Not the Algorithm: Dissociating Answer-Aligned Logit Structure from Embedding Fourier Consolidation in Grokking

Anonymous authors

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## Abstract

The circular Fourier ring that appears in PCA of a grokked transformer’s token embeddings is an iconic visualization in the grokking literature, and is often taken as evidence of the algorithm the network has acquired. We show that its timing runs the other way: the ring is a *lagging* indicator. Measuring three event times within the same runs of a 2-layer modular-arithmetic transformer under decoder-only weight decay—answer-aligned logit Fourier alignment  $\tau_{\text{circuit}}$ , behavioral generalization  $\tau_{\text{gen}}$ , and embedding-Fourier consolidation  $\tau_F$ , each a BIC changepoint—we find  $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$ , with the visible ring consolidating a median  $\sim 6,000$  steps *after* the network already generalises. Three results establish that the embedding-ring lag is functional rather than cosmetic. **(i)** Time-resolved interventions locate the functional dependence: at checkpoints between  $\tau_{\text{circuit}}$  and  $\tau_F$ , removing the dominant answer-aligned Fourier subspace from the logits drops test accuracy by up to 0.72 while matched-energy random removals are negligible (specificity  $\geq 12\times$ ), and a label-free internal patching pilot reproduces this specificity inside the network at the final residual stream (median 0.49 vs 0.02). **(ii)** The three-event ordering holds in 48/54 runs, and in 44/55 when the same design is re-trained on different GPU/CUDA hardware. **(iii)** The lag itself is robust:  $G < F$  in 89.8% of 236 Grokking runs spanning two tasks, two primes, and a dense (lr, wd) grid, and a targeted slow-grokking sweep (24/24  $G < F$  at  $\tau_{\text{gen}}$  up to 54,500 steps) closes off the leading alternative explanation, that the lag merely tracks generalization speed. This sharpens Liu et al. (2022)’s observation that grokking “coincides with the emergence of structure in the embeddings” into a measured within-run lag, and complements Nanda et al. (2023)’s circuit-formation account by isolating the visible ring as a distinct, later event. Practitioners reading embedding geometry as a progress measure should therefore expect it to *understate* how far the computation has already advanced.

## 1 Introduction

Grokking—the abrupt onset of generalization long after training accuracy saturates—has become a canonical testbed for studying delayed representation learning in neural networks (Power et al., 2022). Subsequent work has both extended grokking far beyond toy modular arithmetic, documenting it on broader architectures and datasets (Liu et al., 2023; Humayun et al., 2024; Miller et al., 2024), and developed mechanistic and theoretical accounts of *when* and *why* the transition occurs (Nanda et al., 2023; Varma et al., 2023; Lyu et al., 2024; Rubin et al., 2024; Kumar et al., 2024; Mohamadi et al., 2024). On modular arithmetic tasks, Nanda et al. (2023) showed that grokked transformers implement a Fourier circuit: the network uses trigonometric identities over a small set of task-relevant harmonics to compute modular addition. As a consequence, the  $p$  token embeddings form a near-circular ring in PCA space, with embedding  $i$  at angle  $2\pi i/p$ . That visualization is compelling, but it can also conflate two questions: whether Fourier structure is causally used for the computation, and whether the *embedding table itself* has already settled into a Fourier geometry.

Nanda et al. (2023) also introduced mechanism-level *progress measures*—quantities such as the Fourier-squared loss—and found that these measures improve during a circuit-formation phase that *precedes* the generalization

jump. This raises a natural question about a related but distinct quantity: the token *embedding geometry* itself. Does embedding-level Fourier alignment also consolidate before generalization, or does it lag behind?

Answering this requires measuring the relative timing of two events:

- $\tau_{\text{gen}}$ : generalization onset—the first checkpoint where test accuracy  $\geq 0.9$  and remains there for three consecutive evaluations;
- $\tau_F$ : embedding-geometry consolidation—the BIC changepoint at which token-embedding Fourier alignment rises above a permutation-null baseline (changepoint on null-corrected Fourier  $R^2$ ); a consolidation changepoint rather than a first-appearance time (Section 6.6, E7).

Prior work has not directly measured  $\Delta\tau = \tau_F - \tau_{\text{gen}}$ . Nanda et al. (2023) studied mechanism-level dynamics but did not separately track embedding geometry. Liu et al. (2022) observed that grokking “coincides with the emergence of structure in the embeddings”, but did not measure within-run event timing. Our work closes this gap by separating *when* the answer-aligned Fourier structure becomes functionally necessary from *when* the visible Fourier geometry consolidates.

### Contributions.

1. **We localise where the Fourier computation is functionally load-bearing at the moment of generalization, and it is not the visible ring.** Between  $\tau_{\text{circuit}}$  and  $\tau_F$ , removing the dominant top-3 answer-aligned Fourier harmonics from the logits drops test accuracy by 0.40–0.72 across the eight addition-mod-53 cells that fully grok, while matched-energy random subspace removals drop accuracy by 0.000–0.013 (specificity ratios 12–214 $\times$ ). A top- $K$  ( $K \in \{1, 3, 5, 10\}$ ) ablation spectrum and a frequency-selectivity control place the effect on high-power low-order harmonics specifically, and a label-free activation-patching pilot reproduces this specificity inside the network, at the second-layer components and the final residual stream (median drop 0.49 vs 0.02 for a dimension-matched control; Section 6.6, E7). The causal evidence covers addition mod 53 and mod 97; under discrete-log reindexing of multiplication the same subspace carries no measurable load, which locates the boundary of the localization claim (E7).
2. **A three-event timing framework makes the dissociation measurable within a single run.** Computing logit-level Fourier alignment  $F_L$  alongside the embedding-level  $F_{\text{corr}}$  at every checkpoint recovers  $(\tau_{\text{circuit}}, \tau_{\text{gen}}, \tau_F)$  from one trajectory via BIC changepoint detection on null-corrected Fourier alignment, with a permutation null that calibrates each event against chance alignment. The ordering  $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$  holds in 48/54 runs (88.9%, 95% CI 77.8–94.8%) and in 44/55 when the same design is re-trained on different GPU/CUDA hardware (80.0%, inside that interval; Section 6.6, E3). The circuit-side event is proxy-defined by construction:  $F_L$ , excluded loss and restricted loss (Nanda et al., 2023) agree only at rank level (Spearman  $\rho=0.42$ –0.54) and place  $\tau_{\text{circuit}}$  at different steps, so we read the three detectors as measuring different facets of circuit formation rather than one canonical event (E3). The embedding-ring result in contribution 3 does not depend on this choice, since it is defined by  $\tau_{\text{gen}}$  and  $\tau_F$  alone.
3. **The embedding-ring lag holds across every regime we can reach.** Embedding-geometry consolidation lags generalization in 212/236 Grokking runs (89.8%, CI 85.3–93.1%), spanning 1D weight-decay and learning-rate sweeps (52/55, CI 85.1–98.1%), a dense 2D (lr  $\times$  wd) grid (109/130, CI 76.6–89.2%), a different prime  $p=97$  (21/21, CI 84.5–100%), and modular multiplication via discrete-log reindexing (30/30, CI 88.6–100%). A targeted slow-grokking sweep (24/24 G<F at  $\tau_{\text{gen}}$  up to 54,500 steps) closes off the leading alternative account—that the lag merely tracks generalization speed—and a 36-configuration detector-sensitivity study (mean robustness 0.94; no ordering flips when  $|\Delta\tau| \geq 3,000$  steps) shows the result is not produced by the detector’s hyperparameters.

The core claim is the embedding-ring lag itself: the ring is a post-generalization consolidation signal, and the structure it depicts is already functionally load-bearing well before the picture becomes legible. That

claim rests on  $\tau_{\text{gen}}$  and  $\tau_F$  together with the interventions, and is independent of how the circuit-side event is operationalised; the three-event framework of contribution 2 adds resolution to the picture rather than carrying it. We establish this for the standard 2-layer modular-arithmetic transformer under decoder-only weight decay, the setting in which the ring visualization is normally reported; scope conditions, including a depth pilot in which the ordering reverses, are stated in Section 6.7.

## 2 Background

### 2.1 Grokking and the Phase Diagram

Power et al. (2022) documented grokking in small transformers on binary operations including  $(a + b) \bmod p$ . Liu et al. (2022) mapped training behavior across a (decoder learning rate, weight decay) grid, identifying four phases: *Grokking* (delayed generalization), *Memorization* (no generalization), *Comprehension* (immediate generalization), and *Confusion* (no learning). Our experiments follow their architecture and phase conventions.

### 2.2 Fourier Circuits and Embedding Geometry

Nanda et al. (2023) reverse-engineered the algorithm underlying grokked transformers: the network computes modular sums via trigonometric identities over task-relevant Fourier harmonics. They introduced mechanism-level progress measures and found that circuit formation—the process by which the model constructs the Fourier computation—*precedes* the test-accuracy jump, followed by a “cleanup” phase in which memorization components are removed. Their analysis focused on the computational mechanism (logit-level Fourier structure and progress measures); we study *both* that mechanism and a complementary quantity: the geometric organization of the *token embeddings themselves*. We introduce a logit Fourier alignment score  $F_L$  that measures circuit-level structure (how much logit variance is explained by Fourier harmonics of the answer class), and compare its onset  $\tau_{\text{circuit}}$  with the embedding-geometry consolidation changepoint  $\tau_F$  in the same runs. Progress measures /  $F_L$  track how much of the Fourier *computation* is already in place;  $F_{\text{corr}}$  tracks when the token *embedding table* has converged to its final geometry.

## 3 Related Work

**Grokking and delayed generalization.** Power et al. (2022) introduced grokking on modular arithmetic. Liu et al. (2022) constructed the (lr, wd) phase diagram we use as our experimental scaffold, identifying the four training phases we adopt. Liu et al. (2023) showed that grokking extends well beyond algorithmic tasks and is governed by the ratio of initial weight norm to the norm of the generalising solution; Humayun et al. (2024) subsequently argued that grokking-like delayed-generalization dynamics are pervasive even in standard classification networks, and Miller et al. (2024) document the same phenomenon outside neural networks entirely, suggesting the dynamics studied here are an instance of a broader class. Thilak et al. (2022) linked grokking to cyclic instabilities of adaptive optimisers, terming this the “slingshot mechanism.” On the theoretical side, Rubin et al. (2024) model grokking as a first-order phase transition, Lyu et al. (2024) prove that a dichotomy of early- and late-phase implicit biases can provably induce grokking in homogeneous networks, and Mohamadi et al. (2024) give a margin-based theoretical analysis specific to modular addition—directly motivating our use of modular addition as the primary testbed. Kumar et al. (2024) reframe grokking as the transition from “lazy” to “rich” (feature-learning) training dynamics, which provides a candidate mechanism for the long delay between  $\tau_{\text{gen}}$  and  $\tau_F$  that we measure. Doshi et al. (2024) disentangle generalization from memorization on corrupted modular-arithmetic data, finding that the two events follow distinct trajectories controlled by different hyperparameters; this complements our finding that the logit-level Fourier proxy for circuit formation, generalization, and embedding-geometry consolidation can be temporally pried apart within a single uncorrupted training run. These works characterise *when* or *whether* generalization occurs; our work characterises its timing *relative* to embedding geometry and a logit-level Fourier alignment proxy for circuit formation in the same runs.

**Mechanistic interpretability of grokked models.** The circuit-level view of neural networks (Olah et al., 2020; Elhage et al., 2021) treats trained models as collections of identifiable computational subgraphs;

modular arithmetic has become a flagship setting because the circuit can be fully reverse-engineered. Nanda et al. (2023) reverse-engineered the Fourier circuit and showed via progress measures that circuit formation precedes the generalization jump. Our  $F_L$  metric is analogous to their restricted logit loss but computed for all  $(a, b)$  pairs simultaneously, providing a single continuous signal for BIC changepoint estimation of  $\tau_{\text{circuit}}$ . Varma et al. (2023) argue that grokking reflects the delayed ascent of an efficient generalising circuit over a more immediately learnable memorising solution; Merrill et al. (2023) make this competition concrete by characterising it as a contest between sparse and dense subnetworks, in which the sparse generalising circuit eventually wins out. This sparse-vs-dense framing predicts a temporal gap between *circuit assembly* and *circuit dominance*, consistent with the C<G separation we measure. Chughtai et al. (2023) study how networks learn group operations more broadly, finding that the learned circuits depend on the group structure in a predictable way. Zhong et al. (2023) show that modular addition admits two distinct algorithms (the “Clock” and “Pizza” variants), depending on initialisation and architecture, revealing that the Fourier circuit is not the unique generalising solution. Gromov (2023) provides analytic weight solutions for two-layer networks on modular arithmetic tasks, establishing complete interpretability of the learned representations. Conmy et al. (2023) automate circuit discovery via activation patching, complementing our temporal approach to identifying when circuits form. Beyond modular arithmetic, the circuit framework has matured to larger models: induction heads in language models (Olsson et al., 2022), the indirect-object-identification circuit in GPT-2 (Wang et al., 2023), and sparse-autoencoder feature dictionaries (Bricken et al., 2023; Huben et al., 2024) are recent examples of mechanistic explanation at scale. Our  $F_{\text{corr}}$  metric is complementary to all of the above: it tracks when the token *embedding table* reaches its final geometric arrangement, which lags generalization by thousands of steps in most configurations.

**Implicit bias and weight decay.** Our use of weight decay as the principal axis of variation echoes a long line of work on the implicit biases that accompany gradient descent. Soudry et al. (2018) proved that gradient descent on exponentially-tailed losses converges to the max-margin solution, identifying a direction-selection bias that operates on a much slower timescale than the loss decrease itself. D’Angelo et al. (2024) examine why explicit weight decay remains necessary in modern transformer training despite the use of adaptive optimisers, and Galanti et al. (2022) prove that SGD with weight decay biases learned representations toward low effective rank. The G<F lag we observe (thousands of steps between generalization and the BIC-detected consolidation of Fourier-aligned embedding geometry) fits naturally with this picture: once the generalising circuit is in place, decoder-side weight decay continues to reshape the embedding rows indirectly through back-propagated gradients (since  $\text{wd}_{\text{emb}}=0$  in our setting), producing a measurable geometric reorganisation that postdates  $\tau_{\text{gen}}$ .

**Embedding geometry and representation dynamics.** Saxe et al. (2014) showed that learning in deep linear networks proceeds in discrete stages corresponding to singular value decomposition, an early theoretical account of structured representation formation. Pappayan et al. (2020) identified “Neural Collapse,” the terminal convergence of within-class activations to a simplex equiangular tight frame, as a generic post-convergence geometric phenomenon in classification networks. Zhu et al. (2021) prove that Neural Collapse is the unique global optimum of an unconstrained-features model with weight decay, and Kothapalli (2023) survey the broader empirical and theoretical scope of the phenomenon. Our  $\tau_F$  result is analogous to Neural Collapse in spirit but distinct in content: the Fourier ring geometry of token embeddings is also a *post-generalization* geometric consolidation rather than a prerequisite for generalization, but the converged target is the cyclic-group representation rather than a simplex equiangular tight frame. Elhage et al. (2022) show that polysemanticity and superposition are widespread in transformer features; our focus on the token embedding’s one-dimensional ring structure provides a concrete, measurable geometric ground truth against which timing can be assessed. Barak et al. (2022) demonstrate that abrupt generalization can hide gradual progress in loss-scale metrics, motivating our BIC changepoint approach to detect latent structural transitions.

**Algorithmic learning in transformers.** Modular addition is one of several algorithmic tasks where small transformers have been studied as testbeds for mechanistic interpretation. Lee et al. (2024) examine how data formatting and curriculum choices govern whether small transformers learn integer arithmetic at all; Quirke & Barez (2024) give a circuit-level account of two-digit addition; and Charton (2023) interprets the representations a transformer uses to compute the greatest common divisor. These works confirm that

algorithmic tasks share recurring circuit-formation motifs, suggesting our temporal three-stage framework—measured here on  $(a + b) \bmod p$  and  $(a \cdot b) \bmod p$ —is a plausible candidate for application to those settings.

**Methods.** We use AdamW (Loshchilov & Hutter, 2019) and adopt the transformer architecture of Vaswani et al. (2017) as extended to the circuit framework of Elhage et al. (2021). Our changepoint estimator is based on the Bayesian Information Criterion (Schwarz, 1978); Truong et al. (2020) survey alternative offline changepoint detectors, and the PELT algorithm of Killick et al. (2012) provides a more general optimisation procedure than the single-break search we adopt here. We chose the simpler 1-break formulation because it matches the qualitative shape of  $F_{\text{corr}}$  (a flat baseline followed by a single rising segment) and admits an interpretable persistence check via the post-break slope. A 2-break sanity check on the same 8 representative cells (Section 6.6, “Detector-specification sensitivity”) shows that allowing a second breakpoint does add a late saturation knee that BIC slightly prefers, but leaves the early breakpoint within median 1,500 steps of the canonical 1-break  $\tau_F$ ; we view PELT-style multi-break estimation as a refinement that resolves the saturation knee directly rather than as a correction to the timing claim.

**Alternative geometric alignment metrics.** Several recent works diagnose representation geometry on algorithmic tasks through group-representation theory rather than through a Fourier projection on the answer class. Chughtai et al. (2023) cast modular-arithmetic solutions in terms of irreducible representations of the cyclic group, showing that the network learns specific irreps and that universality across architectures is partly explained at the group-theoretic level. Zhong et al. (2023) identify a complementary “pizza” representation alongside the canonical clock, and show that the choice between the two depends on architectural details of the model. Our  $F_{\text{corr}}$  and  $F_L$  are best read as cheap, single-harmonic proxies for the dominant cyclic irrep on  $(a + b) \bmod p$ : they trade the generality of full group-representation diagnostics for a scalar timing signal that the BIC pipeline can locate as a single changepoint. The three-stage timing claim in this paper does not rely on  $F_{\text{corr}}$  being the optimal geometric metric; any monotone geometric alignment score on the cyclic-group representation that crosses a stable threshold should yield a  $\tau_F$  within measurement resolution of ours, and we expect group-representation diagnostics in the style of Chughtai et al. (2023) to give a compatible but architecture-aware refinement.

**Concurrent and adjacent work.** Several recent papers study representation-level dynamics in modular-arithmetic grokking and partially overlap with our measurement framework, but each addresses a different question. Khaliq (2026) characterises a post-grokking “scaffold” phase in a 2-layer embedding MLP and argues that the post-collapse embedding harmonic structure causally constrains attractor selection. The two results address different windows and compose: our claim is about *the moment of generalization*, where the load-bearing Fourier structure sits in logit space rather than in the visible ring, while theirs is about the post- $\tau_F$  regime, where consolidated embedding structure can shape which grokked attractor the system settles into. Truong Xuan Khanh et al. (2026) report that representation-covariance spectral entropy collapses *before* generalization and couples to the emergence of Fourier-aligned representations; their observable (covariance spectral entropy) differs from our single-harmonic answer-class logit-Fourier subspace, and they do not separate logit-side from embedding-side timing. Xu (2026) use commutator-defect/curvature precursors that precede generalization across learning-rate regimes including a slow regime; their precursor is geometric in weight space and their causal interventions act on gradient flow, not Fourier subspaces. AlquBoj et al. (2025) identify embedding update dynamics and bilinear coupling as grokking mechanisms but do not measure embedding-Fourier alignment timing. Swaroop (2026) analyses ReLU-MLP modular addition and argues that the algorithmic structure is substantially encoded during memorization, an MLP analog of our “logit-Fourier computation forms before generalization” claim. Wang et al. (2026) localise grokking transitions using spectral and Wasserstein diagnostics on representation covariance; their diagnostics target the *generalization transition* as a single event rather than separating logit-side computation from embedding-side geometry, so their measurement is complementary to the three-event decomposition  $(\tau_{\text{circuit}}, \tau_{\text{gen}}, \tau_F)$  we develop here.

**Positioning.** Table 1 summarises what each of the most closely related works measures. To our knowledge, ours is the first study to measure all three as changepoint-detected events in the same runs and pair them with a time-resolved logit-Fourier intervention that distinguishes the load-bearing logit-Fourier subspace from the visible embedding ring.



Table 1: What prior work measures on grokked modular-arithmetic transformers, relative to the three-stage picture. “✓” = directly measured and reported as a timing event; “~” = qualitatively discussed but not measured as an onset time; “.” = not considered. To our knowledge, ours is the first to measure all three in the same runs.

| Work                | $\tau_{\text{circuit}}$ (circuit) | $\tau_{\text{gen}}$ (generalization) | $\tau_F$ (embedding geometry) |
|---------------------|-----------------------------------|--------------------------------------|-------------------------------|
| Power et al. (2022) | .                                 | ✓                                    | .                             |
| Liu et al. (2022)   | .                                 | ✓                                    | .                             |
| Nanda et al. (2023) | ✓                                 | ✓                                    | ~                             |
| Varma et al. (2023) | ~                                 | ✓                                    | .                             |
| Liu et al. (2023)   | .                                 | ✓                                    | ~                             |
| Rubin et al. (2024) | .                                 | ✓                                    | .                             |
| Lyu et al. (2024)   | .                                 | ✓                                    | .                             |
| Doshi et al. (2024) | .                                 | ✓                                    | .                             |
| <b>This work</b>    | ✓                                 | ✓                                    | ✓                             |

## 4 Methods

### 4.1 Task and Model

We train a decoder-only transformer (Vaswani et al., 2017) on  $(a + b) \bmod 53$ . All  $p^2 = 2,809$  input pairs are generated; 30% (843 pairs) form the training set. The model uses  $d_{\text{model}} = 256$ , 4 attention heads, 2 layers, feed-forward dimension 1024, no dropout. The embedding layer has fixed learning rate  $\eta_{\text{emb}} = 10^{-3}$  and weight decay 0; only decoder parameters are swept. We use AdamW (Loshchilov & Hutter, 2019) with a 10-step linear warmup followed by constant learning rate, and gradient clipping at norm 1.0.

### 4.2 Metrics

**Generalization onset  $\tau_{\text{gen}}$ .** Test accuracy is evaluated every 500 steps.  $\tau_{\text{gen}}$  is the first checkpoint where test accuracy  $\geq 0.9$  for three consecutive evaluations (1,500 steps of sustained performance).

**Fourier alignment score  $F(t)$ .** Let  $E_t \in \mathbb{R}^{p \times d}$  be the (mean-row-centered) token embedding matrix at step  $t$ . We compute the fraction of variance explained by the best single Fourier harmonic:

$$F_{\text{raw}}(t) = \max_{k=1}^{\lfloor p/2 \rfloor} \left( 1 - \frac{\|E_t - Q_k Q_k^\top E_t\|_F^2}{\|E_t\|_F^2} \right), \quad (1)$$

where  $Q_k$  is an orthonormal basis for  $[\cos(2\pi k i/p), \sin(2\pi k i/p)]_{i=0}^{p-1}$ .

**Null correction.** With  $p = 53$  and  $d = 256$ , any embedding matrix achieves non-trivial  $F_{\text{raw}}$  by chance. Unless otherwise stated, the canonical detector refreshes the permutation null at every 500-step log point, matching the metric-logging cadence; for sensitivity analysis we also evaluate refresh intervals of 2,500 and 5,000 steps (Section 6.5). At each refresh step we estimate the 95th percentile of  $F_{\text{raw}}$  under 100 random token-index permutations of the *current* embedding matrix:

$$F_{\text{null,p95}}(t) = \text{Pct}_{95} \left( \{F_{\text{raw}}(E_t[\sigma]) \mid \sigma \in \mathcal{S}^{(100)}\} \right), \quad (2)$$

and set  $F_{\text{corr}}(t) = \max(0, F_{\text{raw}}(t) - F_{\text{null,p95}}(t))$ .

**Null trajectory between updates.**  $F_{\text{raw}}$  is evaluated at every 500-step log point, and the canonical detector configuration refreshes the permutation null  $F_{\text{null,p95}}$  at the same cadence (every 500 training steps; see Appendix A for the configuration table). Between refreshes we use a last-observation-carried-forward (LOCF) schedule: the null value is held at its most recent estimate and *no interpolation* is applied. LOCF is a deliberate choice: it is causal (uses no future embeddings), matches an on-line detector that could be

deployed to monitor the same signal during training, and avoids leakage of future null estimates into earlier  $F_{\text{corr}}$  values. All downstream processing (EMA smoothing at  $\alpha=0.15$  and BIC-selected segmented regression) operates on  $F_{\text{corr}} = \max(0, F_{\text{raw}} - F_{\text{null,p95}}^{\text{LOCF}})$ , so any step discontinuity in the null trajectory at a refresh boundary propagates into  $F_{\text{corr}}$  but is partially damped by the EMA. At the canonical 500-step refresh cadence, refresh boundaries coincide with measurement boundaries, so the LOCF segment length collapses to a single log point and the worst-case hold-forward displacement that LOCF could induce is bounded above by one measurement bin.

**Artefact check near null-refresh boundaries.** The LOCF artefact only becomes visible if the refresh cadence is loosened beyond the measurement bin: an LOCF null with refresh interval  $R$  could in principle bias  $\tau_F$  by up to  $R$  steps if the changepoint happens to fall close to a refresh step. We therefore treat the null-update cadence as a free detector hyperparameter and sweep it jointly with the EMA coefficient and the BIC slope threshold: null interval  $\in \{500, 2500, 5000\}$  (i.e.  $1\times, 5\times, 10\times$  the canonical bin),  $\alpha \in \{0.1, 0.15, 0.3, 1.0\}$ , slope threshold  $\in \{0.005, 0.01, 0.02\}$  (36 configurations; Section 6.5). Varying the refresh cadence by a factor of 10, the dimension directly probing the LOCF artefact, produces  $\tau_F$  shifts within the 500-step measurement-resolution floor and does not change any ordering label for the 7 Grokking cells tested: five cells with  $|\Delta\tau| \geq 5,000$  steps remain G<F in all 36/36 configurations, and no G<F cell is ever reclassified as F<G under any configuration. We conclude that the piecewise-constant null trajectory does not produce a detectable changepoint artefact at the 500-step resolution at which  $\tau_F$  is reported.

**Fourier geometry consolidation changepoint  $\tau_F$ .** We apply a BIC-selected 1-break segmented regression (Schwarz, 1978) to  $F_{\text{corr}}(t)$  on a log-time scale after exponential-moving-average smoothing ( $\alpha=0.15$ ). The choice of log-time over linear-time segmentation is motivated by the geometry of grokking trajectories: the changepoints of interest ( $\tau_{\text{gen}}, \tau_F, \tau_{\text{circuit}}$ ) all lie in the early fraction of the run, and on a 50,000-step linear axis the late plateau dominates the regression target so that a 1-break fit is biased toward the right of the trajectory. A log-time axis equalises the density of regression points across the orders of magnitude over which the changepoints actually move (roughly  $10^3$  to  $10^4$  steps), giving the BIC criterion comparable evidence in early and late segments. A changepoint is accepted only if the 4-parameter model strictly improves BIC over the 2-parameter no-break baseline and the post-break slope exceeds 1% of the total value range.  $\tau_F$  is the accepted breakpoint step; “not detected” otherwise.

**BIC penalty, time discretisation, and error model.** The segmented regression is fit on log-discretised time  $x_t = \log(1 + t)$  where  $t \in \{500, 1000, \dots, T_{\text{max}}\}$  is the training step at which  $F_{\text{corr}}$  is logged. For a candidate model with  $k$  free parameters (no-break,  $k=2$ ; 1-break,  $k=4$ ) and residual sum of squares RSS over  $n$  log-time samples, the BIC score is  $\text{BIC} = n \ln(\text{RSS}/n) + k \ln n$  (Schwarz, 1978), i.e. the Schwarz form with the i.i.d.-Gaussian residual likelihood constant absorbed. **The canonical detector compares the no-break model against the 1-break model only.** The 2-break model is not used for headline  $\tau_F$  and is analysed only post-hoc (Section 6.6, “Detector-specification sensitivity”). The canonical rule is therefore: accept the 1-break breakpoint if its BIC strictly improves on the no-break baseline, subject to the post-break slope and minimum-segment constraints described above; otherwise return “not detected”. The estimator assumes residuals of  $F_{\text{corr}}(x) - \hat{f}(x)$  are i.i.d. Gaussian on the log-time axis, and the 36-configuration sensitivity study of Section 6.5 establishes that the recovered ordering is stable across the detector settings this assumption interacts with.

**Logit Fourier alignment score  $F_L(t)$  and circuit onset  $\tau_{\text{circuit}}$ .** Let  $\mathbf{L}_t \in \mathbb{R}^{p^2 \times p}$  be the full logit matrix at step  $t$ , where row  $(ap + b)$  is the output logit vector for input pair  $(a, b)$ . Let  $s_{ab} = (a + b) \bmod p$  be the target class. Column-centering  $\mathbf{L}_t^c = \mathbf{L}_t - \bar{\mathbf{L}}_t$ , we compute:

$$F_L(t) = \max_{k=1}^{\lfloor p/2 \rfloor} \left( 1 - \frac{\|\mathbf{L}_t^c - Q_k^{(s)} Q_k^{(s)\top} \mathbf{L}_t^c\|_F^2}{\|\mathbf{L}_t^c\|_F^2} \right), \quad (3)$$

where  $Q_k^{(s)}$  is an orthonormal basis for  $[\cos(2\pi k s_{ab}/p), \sin(2\pi k s_{ab}/p)]$  indexed over all  $p^2$  input pairs. This measures the fraction of total logit variance explained by the best single Fourier harmonic of the answer

class, analogous to Nanda et al.’s restricted logit loss. A permutation-null-corrected version  $F_L^{\text{corr}}$  is computed identically to  $F_{\text{corr}}$  (permuting the  $s_{ab}$  labels).  $\tau_{\text{circuit}}$  is the BIC changepoint of  $F_L^{\text{corr}}$ , estimated with the same procedure as  $\tau_F$ .

**Interpretation of  $\tau_F$  and  $\tau_{\text{circuit}}$  as single-harmonic changepoints.** Both  $F_{\text{corr}}$  and  $F_L$  take a maximum over *single* Fourier subspaces ( $k = 1, \dots, \lfloor p/2 \rfloor$ ) rather than fitting the full multi-harmonic basis that the grokked circuit ultimately uses. Accordingly,  $\tau_F$  should be read as the *consolidation changepoint* of above-chance single-harmonic *embedding* organisation, not as the first appearance of functional Fourier content: the E7 embedding intervention (Section 6.6, finding 4) shows embedding Fourier directions are functionally necessary well before  $\tau_F$ .  $\tau_{\text{circuit}}$  is likewise the changepoint of above-chance single-harmonic *logit* organisation; each is a lower-bound proxy for the richer geometry and computation the trained network eventually realises. Figure axis labels use the shorthand “onset” for these changepoints. We adopt the single-harmonic form because it admits a clean permutation null and a single scalar trajectory suitable for BIC changepoint detection, and the multi-harmonic variant of Section 6.6 shows that recomputing  $\tau_F$  at  $K \in \{1, 2, 3, 5\}$  preserves the ordering label of every cell.

**Ordering labels.** Given  $(\tau_{\text{gen}}, \tau_F)$ , each run is labeled: **G<F** ( $\tau_{\text{gen}} < \tau_F$ ); **F<G** ( $\tau_{\text{gen}} > \tau_F$ ); **coincident** ( $|\Delta\tau| \leq 500$ ); **F\_only** ( $\tau_F$  detected,  $\tau_{\text{gen}}$  not); or **none** (neither detected). Three-way ordering among  $(\tau_{\text{circuit}}, \tau_{\text{gen}}, \tau_F)$  is denoted by the corresponding inequality string (e.g., **C<G<F**).

### 4.3 Experimental Design

**Phase classification.** A run is *Grokking* if both  $\tau_{\text{train}}$  and  $\tau_{\text{gen}}$  are detected and  $\tau_{\text{gen}} - \tau_{\text{train}} \geq 2,000$  steps. *Memorization*:  $\tau_{\text{train}}$  detected,  $\tau_{\text{gen}}$  not. *Comprehension*: both detected, gap  $< 2,000$  steps.

**Stage 2 — weight decay sweep.** Fixed lr= $1.6 \times 10^{-3}$ . Weight decay: 10 log-spaced values in  $[1.2, 3.5]$ :  $\{1.20, 1.35, 1.52, 1.71, 1.93, 2.18, 2.45, 2.76, 3.11, 3.50\}$ . Seeds  $\{42, 7, 2025\}$ . Total: 30 runs.

**Stage 3 — learning rate sweep.** Fixed wd=2.5. Learning rate: 10 log-spaced values in  $[5 \times 10^{-4}, 8 \times 10^{-3}]$ :  $\{5.0, 6.8, 9.3, 12.6, 17.1, 23.3, 31.7, 43.2, 58.8, 80.0\} \times 10^{-4}$ . Seeds  $\{42, 7, 2025\}$ . Total: 30 runs.

Training runs up to 50,000 steps, extended to 80,000 if train\_acc  $> 0.9$  but  $\tau_{\text{gen}}$  not yet detected.

**Stage 4 — circuit timing ( $\tau_{\text{circuit}}$ ).** We retrain all Stage 2 and Stage 3 cells (60 runs total), augmenting the training loop with  $F_L$  computed every 500 steps. The same null-correction and BIC changepoint procedure yields  $\tau_{\text{circuit}}$  for each run. Three-way ordering  $(\tau_{\text{circuit}}, \tau_{\text{gen}}, \tau_F)$  is reported.

**Stage 5A — modular multiplication.** Operation:  $(a \cdot b) \bmod 53$ . For the multiplication task, token  $i$  should be indexed by its discrete logarithm  $\log_g(i) \bmod (p-1)$  rather than by  $i$  directly. We reorder embeddings by the discrete log before computing  $F_{\text{raw}}$ , and redefine  $s_{ab} = (a \cdot b) \bmod p$  in  $F_L$ . Fixed lr= $1.6 \times 10^{-3}$ ; wd: 10 log-spaced values in  $[1.2, 3.5]$ ; seeds  $\{42, 7, 2025\}$ ; max 50,000 steps. Total: 30 runs.

**Stage 5B — different prime ( $p=97$ ).** Operation:  $(a + b) \bmod 97$ . Grid: same lr and wd as Stage 2 (lr= $1.6 \times 10^{-3}$ , 10 wd values); max 80,000 steps (larger prime groks slower). Total: 30 runs.

**Stage 6 — dense 2D grid.** Operation:  $(a + b) \bmod 53$ . lr: 7 log-spaced values in  $[5 \times 10^{-4}, 4.3 \times 10^{-3}]$ ; wd: 7 log-spaced values in  $[1.2, 3.5]$ ; seeds  $\{42, 7, 2025\}$ ; max 50,000 steps (auto-extend to 80k). Total: 147 runs.

Code, training configurations, and per-run metric trajectories will be released at <https://github.com/ANONYMISED/three-stage-grokking> (link anonymised for review). Exact commit hashes, random seeds, and package versions used to produce each figure and table are listed in Appendix D.



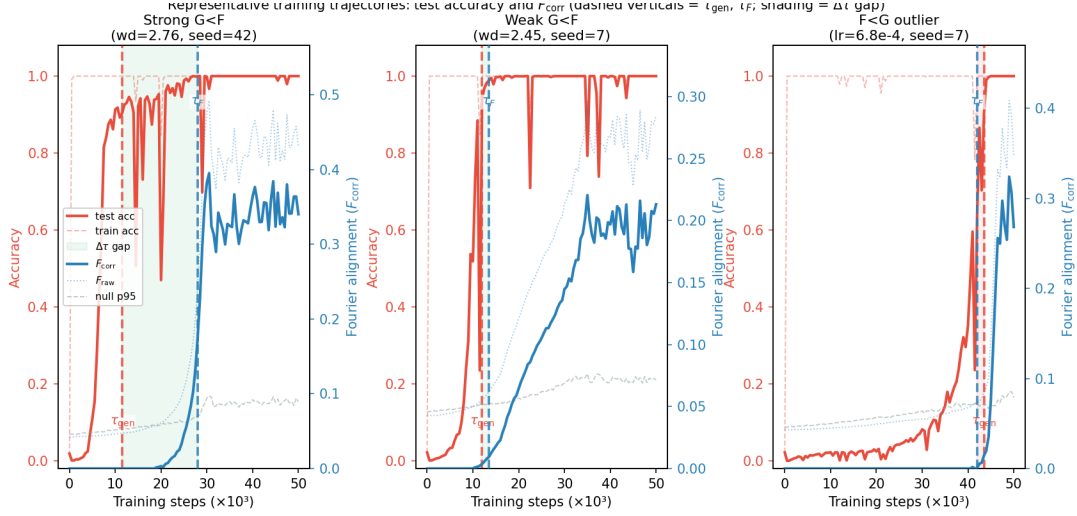


Figure 1: **Example training trajectories for three representative runs.** Red curves: test accuracy (left axis). Blue curves: null-corrected Fourier alignment  $F_{\text{corr}}$  (right axis, dotted = raw  $F_{\text{raw}}$ , grey dashed = null p95 baseline). Dashed verticals mark  $\tau_{\text{gen}}$  (red) and  $\tau_F$  (blue). *Left and centre:*  $G < F$  runs where generalization fires first and the green-shaded  $\Delta\tau$  gap follows;  $F_{\text{corr}}$  rises only after test accuracy has already reached 1. *Right:* the original Stage 3  $F < G$  outlier ( $\text{lr} = 6.8 \times 10^{-4}$ , seed 7), stable under detector hyperparameters but not reproduced in the targeted cross-hardware re-run of Section 7, where  $F_{\text{corr}}$  barely exceeds the null baseline (0.015) one log-interval before an unusually late  $\tau_{\text{gen}} = 43,500$ .

## 5 Results

Figure 1 provides intuition for the two key events and their ordering before we present aggregate statistics.

### 5.1 Stage 2: $G < F$ Across Weight Decay

All 30 Stage 2 runs land in the Grokking phase ( $\text{train\_acc} = 1.0$ ,  $\text{test\_acc} \geq 0.99$ ). **29/30 runs (96.7%, 95% CI 83.3%–99.4%) show  $G < F$  ordering;** the sole exception ( $\text{wd} = 3.11$ , seed 2025) has  $\Delta\tau = 0$  and is labeled coincident (treated as a tie throughout this paper, not counted as  $F < G$ ). **No Stage 2 run exhibits  $F < G$  ordering,** so the headline finding is most precisely stated as “no  $F < G$  within the Grokking phase at the Stage 2 resolution,” with 29/30 strictly  $G < F$  and 1/30 tied. All 95% confidence intervals reported in Results are Wilson intervals for a binomial proportion. Table 2 and Figure 2 summarize the results.  $P(G < F)$  remains near 1.0 across the full  $\text{wd}$  range tested; the median  $\Delta\tau$  fluctuates between 5,500 and 13,000 steps with no monotonic trend, and  $P(F_{\text{only}}) = 0$  throughout.

### 5.2 Stage 3: $G < F$ Across Learning Rate

Of the 30 Stage 3 runs, **25 land in the Grokking phase** for  $\text{lr} \in [5 \times 10^{-4}, 4.3 \times 10^{-3}]$ ; the remaining 5 runs ( $\text{lr} \geq 5.9 \times 10^{-3}$ ) are Memorization. Among the 25 Grokking runs, **23/25 (92.0%, CI 75.0%–97.8%) show  $G < F$  ordering.** Results appear in Table 3 and Figure 3.

The two exceptions to  $G < F$  deserve comment. **(i)**  $\text{lr} = 6.8 \times 10^{-4}$ , seed 7:  $\Delta\tau = -1,500$  steps, above the 500-step resolution floor but spanning only 3 checkpoints; the sensitivity analysis confirms  $F < G$  is stable across all 36 detector configurations (see Section 6.5). **(ii)**  $\text{lr} = 4.32 \times 10^{-3}$ , seed 42:  $\Delta\tau = -54,500$  steps and max test accuracy only 0.965 (required 80k-step extension). This seed sits at the edge of the Grokking phase and had an atypical training trajectory.

Table 2: Stage 2: wd sweep at  $\text{lr} = 1.6 \times 10^{-3}$  (3 seeds per wd value). All 30 runs are Grokking phase.  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  in thousands of steps (positive =  $G < F$ ). Medians computed across seeds.

| wd         | P( $G < F$ ) | med. $\tau_{\text{gen}}$ (k) | med. $\tau_F$ (k) | med. $\Delta\tau$ (k) |
|------------|--------------|------------------------------|-------------------|-----------------------|
| 1.20       | 3/3          | 26.0                         | 34.0              | +7.0                  |
| 1.35       | 3/3          | 20.5                         | 28.5              | +9.0                  |
| 1.52       | 3/3          | 14.5                         | 21.5              | +5.5                  |
| 1.71       | 3/3          | 14.0                         | 26.5              | +6.5                  |
| 1.93       | 3/3          | 14.0                         | 23.5              | +11.0                 |
| 2.18       | 3/3          | 11.0                         | 23.5              | +8.5                  |
| 2.45       | 3/3          | 11.0                         | 16.0              | +5.5                  |
| 2.76       | 3/3          | 10.5                         | 18.5              | +8.5                  |
| 3.11       | 2/3          | 11.0                         | 16.5              | +6.5                  |
| 3.50       | 3/3          | 12.0                         | 21.5              | +9.0                  |
| <b>All</b> | <b>29/30</b> | —                            | —                 | +7.8                  |

Table 3: Stage 3: lr sweep at wd = 2.5 (3 seeds per lr). “Phase” = majority phase across seeds. “—” = Memorization (no  $\tau_{\text{gen}}$  detected).

| lr                    | Phase        | P( $G < F$ ) | med. $\Delta\tau$ (k) | P(F_only) |
|-----------------------|--------------|--------------|-----------------------|-----------|
| $5.0 \times 10^{-4}$  | Grokking     | 3/3          | +2.5                  | 0/3       |
| $6.8 \times 10^{-4}$  | Grokking     | 2/3          | +4.0                  | 0/3       |
| $9.3 \times 10^{-4}$  | Grokking     | 3/3          | +7.5                  | 0/3       |
| $1.26 \times 10^{-3}$ | Grokking     | 3/3          | +9.5                  | 0/3       |
| $1.71 \times 10^{-3}$ | Grokking     | 3/3          | +6.0                  | 0/3       |
| $2.33 \times 10^{-3}$ | Grokking     | 3/3          | +4.0                  | 0/3       |
| $3.17 \times 10^{-3}$ | Grokking     | 3/3          | +6.0                  | 0/3       |
| $4.32 \times 10^{-3}$ | Grokking     | 2/3          | +7.5                  | 0/3       |
| $5.88 \times 10^{-3}$ | Mixed        | 1/1          | +2.5                  | 2/3       |
| $8.00 \times 10^{-3}$ | Memorization | —            | —                     | 3/3       |
| <b>Grokking</b>       | —            | <b>23/25</b> | +4.0                  | 0/25      |

### 5.3 Combined: $\Delta\tau$ Distribution

Combining both sweeps, 55 of 60 runs land in the Grokking phase. Of these, **52/55 (94.5%, CI 85.1%–98.1%) show  $G < F$  ordering**, 2/55 show  $F < G$ , and 1/55 is coincident ( $\Delta\tau = 0$ ). Figure 4 shows the full  $\Delta\tau$  distribution. Positive values ( $G < F$ ) span 1,000–25,500 steps; the two negative values are the outliers noted above. Median  $\Delta\tau \approx 7,000$  steps (IQR  $\approx 4,000$  steps).

### 5.4 Stage 4: Circuit Onset $\tau_{\text{circuit}}$ and Three-Stage Ordering

We retrained all Stage 2+3 cells with the additional  $F_L$  metric, yielding  $\tau_{\text{circuit}}$  for each run.  $\tau_{\text{circuit}}$  was detected in all 54 Grokking runs (100%).

**$C \leq G$  (circuit at or before generalization).** In 49/54 Grokking runs (90.7%),  $\tau_{\text{circuit}} < \tau_{\text{gen}}$  strictly; in one additional run  $\tau_{\text{circuit}} = \tau_{\text{gen}}$  (both detected at the same 500-step checkpoint), giving 50/54 (92.6%, CI 82.4%–97.1%) under the  $C \leq G$  convention used throughout this section. The median gap  $\tau_{\text{gen}} - \tau_{\text{circuit}} \approx 6,250$  steps (range  $-3,000$  to  $35,000$  steps). The 4 exceptions with  $\tau_{\text{circuit}} > \tau_{\text{gen}}$  (i.e.  $G < C < F$ ) occur at the highest learning rates ( $\text{lr} \geq 2.3 \times 10^{-3}$ , wd = 2.5), where generalization is very fast ( $\tau_{\text{gen}} \leq 6,500$  steps) and the logit Fourier signal rises too gradually for BIC to detect a changepoint before  $\tau_{\text{gen}}$ .

**Proxy-level three-stage ordering  $C < G < F$ .** The dominant pattern is  **$C < G < F$**  in 48/54 runs (88.9%, CI 77.8%–94.8%). The remaining runs split into  $C < F < G$  (2 runs, low-lr boundary) and  $G < C < F$  (4 runs,

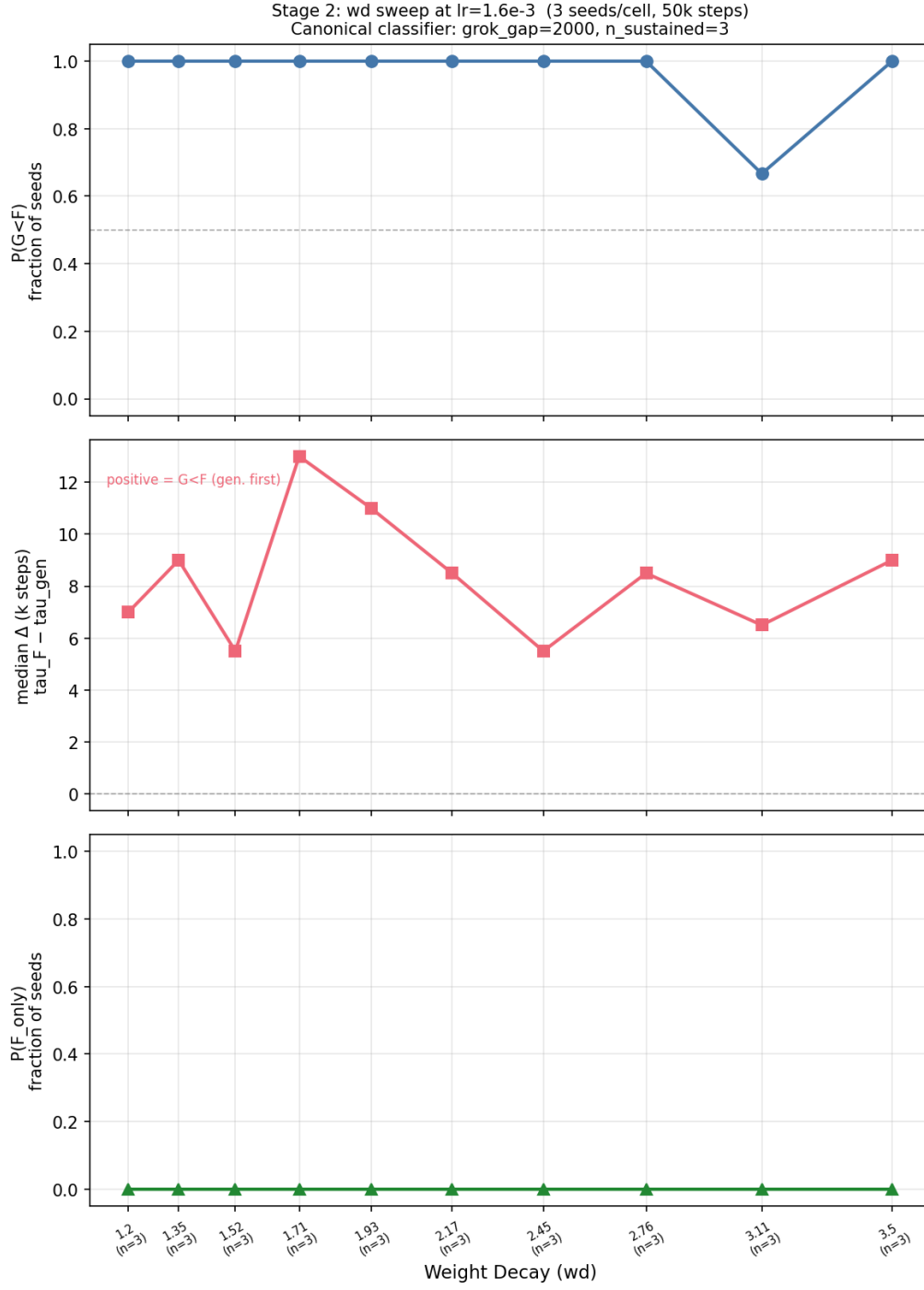


Figure 2: Stage 2: embedding Fourier consolidation vs. generalization across weight decay. **Top:**  $P(G<F)$  per wd value. **Middle:** median  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  (k steps). **Bottom:**  $P(F_{\text{only}})$  (Fourier detected, no generalization). The  $G<F$  plateau is wide: it holds throughout  $\text{wd} \in [1.2, 3.5]$ .

high-lr boundary). The three intervals are nearly equal in magnitude: median  $\tau_{\text{gen}} - \tau_{\text{circuit}} \approx 6,250$  steps, median  $\tau_F - \tau_{\text{gen}} \approx 6,000$  steps, median  $\tau_F - \tau_{\text{circuit}} \approx 13,000$  steps.

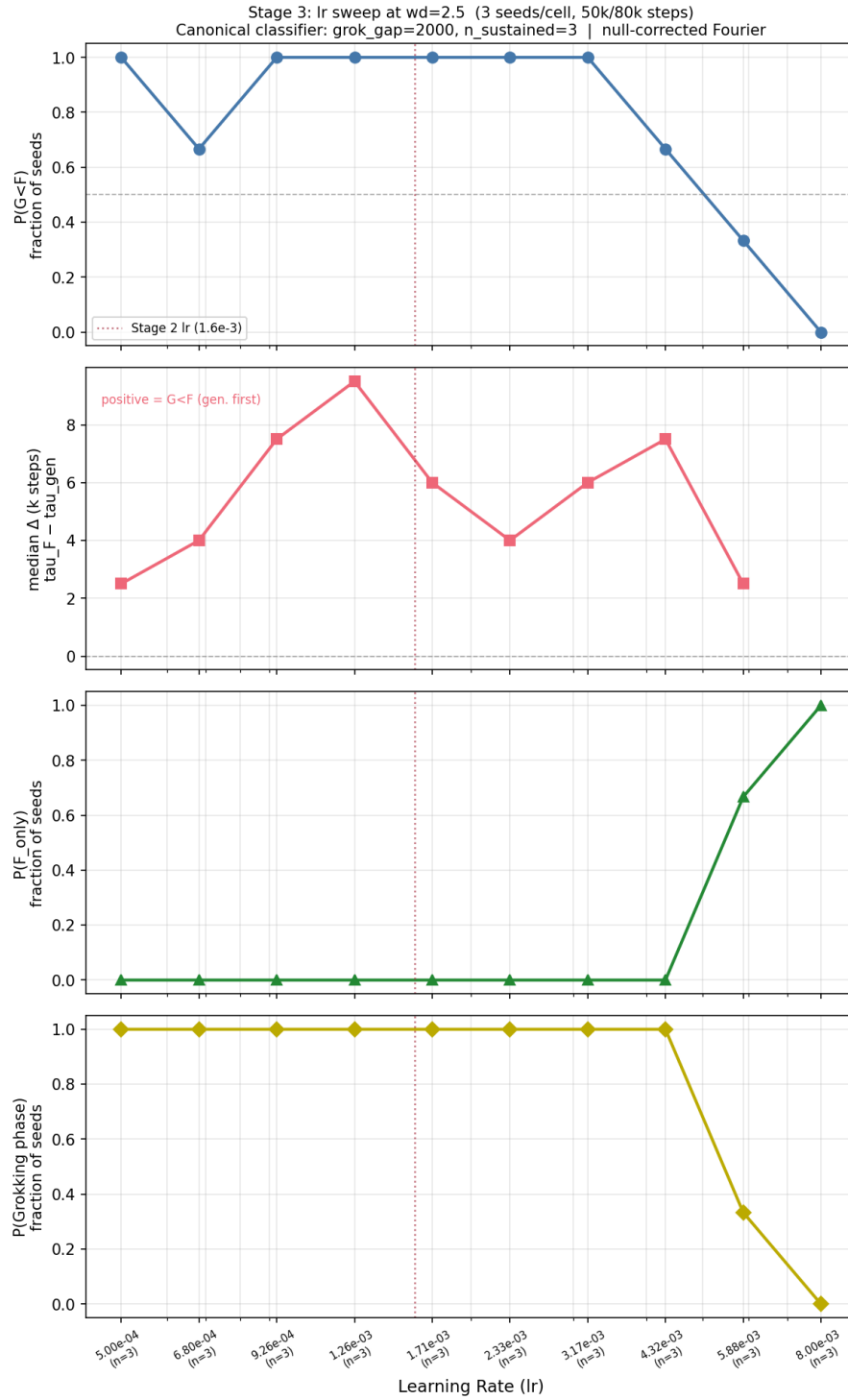


Figure 3: Stage 3: embedding Fourier consolidation vs. generalization across learning rate (wd = 2.5). Layout mirrors Figure 2; x-axis is log-scaled. The bottom panel shows  $P(\text{Grokking})$  per lr: cells at  $\text{lr} \geq 5.9 \times 10^{-3}$  enter Memorization. The vertical dotted line marks the Stage 2 fixed lr ( $1.6 \times 10^{-3}$ ), which lies firmly in the  $G < F$  plateau.

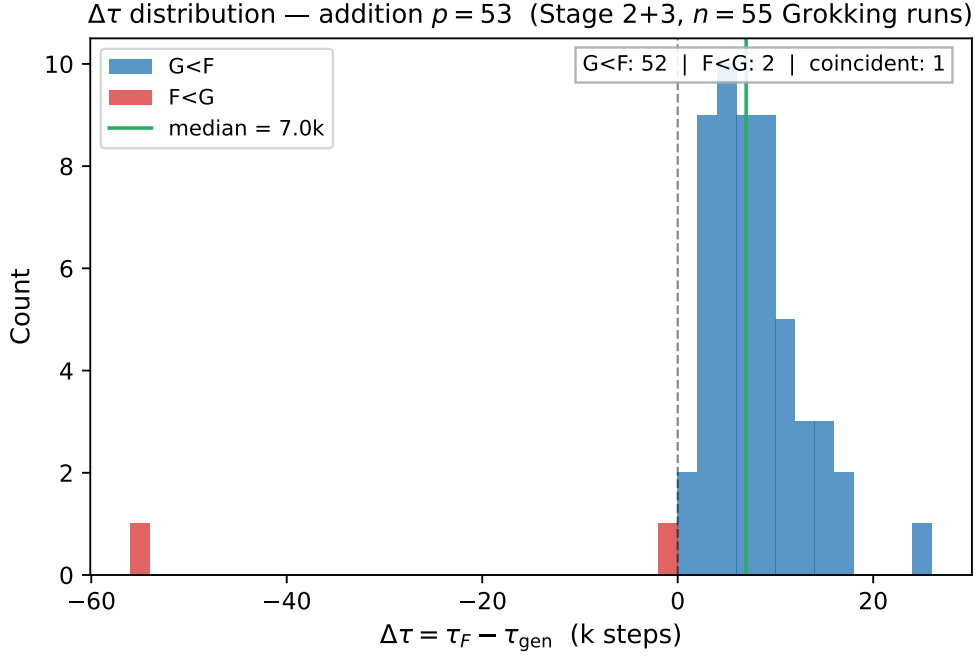


Figure 4: Distribution of  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  across all 55 Grokking runs (Stage 2 blue, Stage 3 red). Positive values indicate  $G < F$  (generalization first). Median  $\Delta\tau \approx 7,000$  steps; 52/55 runs (94.5%) are positive. The two negative values are discussed in Section 5.2.

Table 4: Stage 4: three-stage timing across Stage 2 ( $\text{lr} = 1.6 \times 10^{-3}$ ) and Stage 3 ( $\text{wd} = 2.5$ ) Grokking runs. Medians in thousands of steps.

| Subset             | C<G<F        | C<G<F rate   | med. $\tau_{\text{gen}} - \tau_{\text{circuit}}$ (k) | med. $\tau_F - \tau_{\text{gen}}$ (k) |
|--------------------|--------------|--------------|------------------------------------------------------|---------------------------------------|
| Stage 2 (lr fixed) | 30/30        | 100%         | 6.3                                                  | 6.0                                   |
| Stage 3 (wd fixed) | 18/24        | 75.0%        | 6.3                                                  | 6.0                                   |
| <b>All</b>         | <b>48/54</b> | <b>88.9%</b> | <b>6.3</b>                                           | <b>6.0</b>                            |

## 5.5 Stage 5A: Modular Multiplication

All 30 runs with  $(a \cdot b) \bmod 53$  land in the Grokking phase. **30/30 (100%, CI 88.6%–100%) show  $G < F$  ordering** after discrete-log reindexing of token embeddings. Median  $\Delta\tau = 8,750$  steps (range 3,500–24,000), comparable to the addition results. The perfect  $G < F$  rate extends the finding beyond the addition task, suggesting the ordering is not specific to the  $(a + b) \bmod p$  circuit structure.

**Evaluation-domain check.**  $\tau_{\text{gen}}$  in the Stage 5A headline is computed from the standard 70/30 random split over the full  $p^2$  input pairs (including pairs with a zero operand), whereas  $F_{\text{corr}}$  and  $F_L$  (and hence  $\tau_F$  and  $\tau_{\text{circuit}}$ ) are restricted to the  $\mathbb{F}_p^*$  subgrid (Section 4, “Stage 5A — modular multiplication”). To verify that this evaluation-domain mismatch does not inflate the reported  $G < F$  rate, we re-ran the full 30-cell Stage 5A sweep with identical seeds and additionally tracked test accuracy on the  $\mathbb{F}_p^*$ -only test subset  $(a, b \in \{1, \dots, p-1\})$  at every 500-step checkpoint (`src/sweeps/stage5a_fpstar_recompute.py`; results in `results/stage5a_fpstar/results.csv`). The recompute confirms that the matched-domain  $\tau_{\text{gen}}$  is almost identical to the full-split  $\tau_{\text{gen}}$ :  $\tau_{\text{gen}}^{\mathbb{F}_p^*} - \tau_{\text{gen}}^{\text{full}} \in \{0, 500\}$  steps for all 30 cells (median 0, max 500, i.e. at most one 500-step measurement bin), the median  $\Delta\tau$  is unchanged at 8,750 steps, the minimum  $|\Delta\tau^{\mathbb{F}_p^*}|$  is 3,500 steps, and the 30/30  $G < F$  rate is preserved (no cell flips to coincident or  $F < G$ ). The Stage 5A  $G < F$  claim is therefore not an artefact of the evaluation-domain mismatch; the same caveat does not apply to Stage 5B,



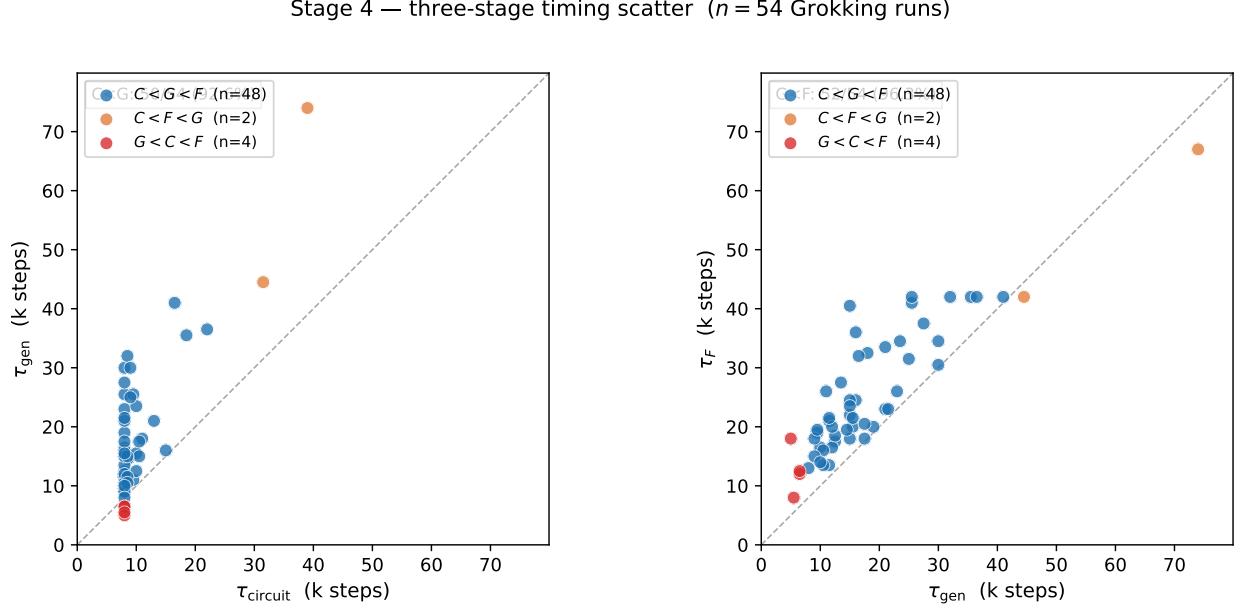


Figure 5: **Three-stage timing scatter (Stage 4).** *Left:*  $\tau_{\text{circuit}}$  (circuit) vs  $\tau_{\text{gen}}$  (generalization). Points above the diagonal ( $y > x$ ) confirm  $C < G$  ordering. Blue:  $C < G < F$  (dominant); orange:  $C < F < G$ ; pink:  $G < C < F$ . *Right:*  $\tau_{\text{gen}}$  vs  $\tau_F$  (embedding geometry). Points above diagonal confirm  $G < F$ .  $C < G$  holds marginally in 50/54 runs (92.6%); the stricter joint  $C < G < F$  pattern in Table 4 requires both  $C < G$  and  $G < F$  in the same run.

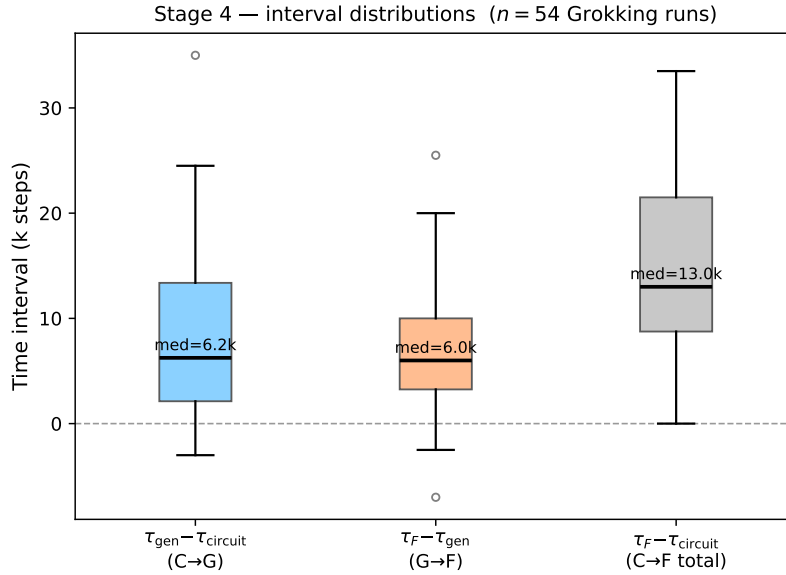


Figure 6: **Three-stage interval distributions (Stage 4,  $n=54$  Grokking runs).** All three intervals are positive on the median, with the total span  $\tau_F - \tau_{\text{circuit}} \approx 13,000$  steps split nearly equally between circuit  $\rightarrow$  generalization and generalization  $\rightarrow$  embedding-geometry.

Stage 2, Stage 3, Stage 4, or Stage 6, all of which use the full-vocabulary additive group  $\mathbb{Z}/p$  and have no analogous trivial subgrid.

Table 5: G<F rates across all experimental stages.  $n_G$ : Grokking runs; G<F: ordering rate and median lag. 95% CIs are Wilson intervals for a binomial proportion. The “All (main stages)” row excludes the targeted slow-grokking sweep of Section 7 (which adds 24/24 G<F cells and brings the aggregate to 236/260=90.8%). A cell-level cluster bootstrap (Table 14) yields 33–40% wider CIs on sources with non-degenerate seed clustering; point estimates shift by <1%.

| Stage                                  | Setting                        | $n_G$      | G<F rate     | 95% CI            | med. $\Delta\tau$ (k) |
|----------------------------------------|--------------------------------|------------|--------------|-------------------|-----------------------|
| 2+3                                    | add mod 53, 1D sweeps          | 55         | 94.5%        | 85.1–98.1%        | 7.0                   |
| 5A                                     | mul mod 53                     | 30         | <b>100%</b>  | 88.6–100%         | 8.8                   |
| 5B                                     | add mod 97                     | 21         | <b>100%</b>  | 84.5–100%         | 28.0                  |
| 6                                      | add mod 53, 2D grid            | 130        | 83.8%        | 76.6–89.2%        | 6.0                   |
| <b>All (main stages)</b>               |                                | <b>236</b> | <b>89.8%</b> | <b>85.3–93.1%</b> | —                     |
| §7                                     | slow-grokking sweep (post-hoc) | 24         | <b>100%</b>  | 86.2–100%         | 11.75                 |
| <b>All (incl. slow-grokking sweep)</b> |                                | <b>260</b> | <b>90.8%</b> | <b>86.6–93.7%</b> | —                     |

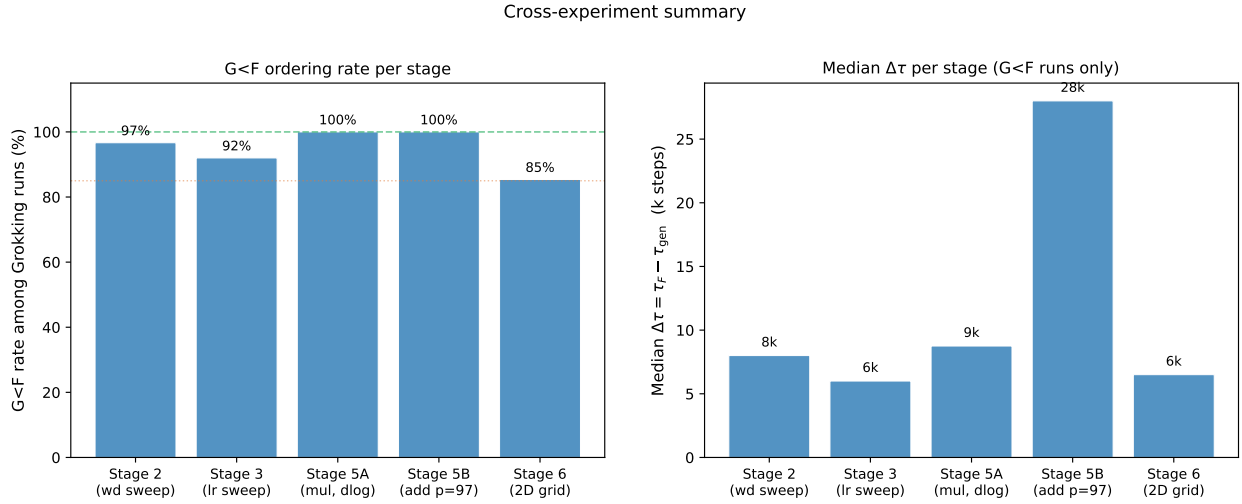


Figure 7: **G<F rate and median lag across all experimental stages.** *Left:* G<F ordering rate; all stages exceed 85%. *Right:* median  $\Delta\tau = \tau_F - \tau_{\text{gen}}$ ; the larger prime  $p=97$  (Stage 5B) shows a markedly larger lag, consistent with slower grokking requiring a longer consolidation period.

## 5.6 Stage 5B: Different Prime ( $p=97$ )

Of the 30 runs with  $(a + b) \bmod 97$ , 21 land in the Grokking phase (9 in Comprehension, concentrated at high wd where fast generalization prevents a Grokking-phase classification). **21/21 Grokking runs (100%, CI 84.5%–100%) show G<F ordering.** The median lag  $\Delta\tau = 28,000$  steps is markedly larger than for  $p=53$  ( $\approx 7,000$  steps), consistent with the larger prime requiring more training steps and a correspondingly longer consolidation period. The G<F rate and direction are unchanged.

## 5.7 Stage 6: G<F Across the Full 2D Phase Diagram

The  $7 \times 7$  grid yields 130 Grokking runs (88.4% of 147 total; 16 Memorization, 1 Comprehension). **109/130 Grokking runs (83.8%, CI 76.6%–89.2%) show G<F ordering** under the canonical  $|\Delta\tau| \leq 500$  coincident threshold; 13/130 show F<G, 3/130 are coincident, and 5/130 have G\_only ( $\tau_F$  not detected). Median  $\Delta\tau \approx 6,000$  steps.

The G<F rate is not uniform across the grid. By learning rate, G<F peaks at intermediate lr ( $\approx 95\%$  for  $\text{lr} \in [10^{-3}, 2 \times 10^{-3}]$ ) and decreases toward both boundaries (58% at  $\text{lr} = 5 \times 10^{-4}$ , 78% at  $\text{lr} = 4.3 \times 10^{-3}$ ).

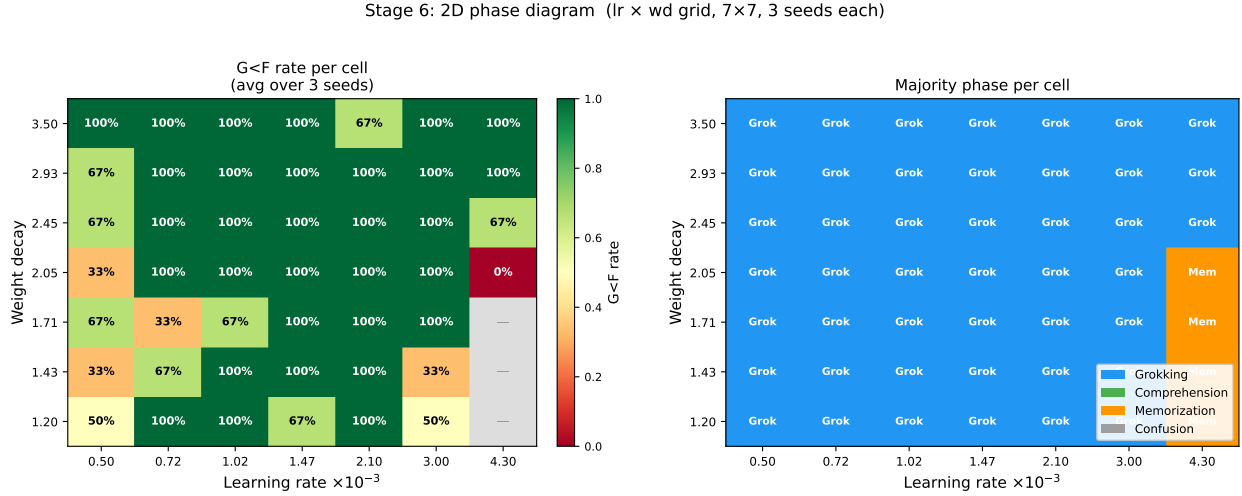


Figure 8: **Stage 6: 2D phase diagram and G<F rate.** *Left:* G<F rate per (lr, wd) cell, averaged over 3 seeds. G<F is strongest in the interior ( $\text{lr} \in [10^{-3}, 2 \times 10^{-3}]$ ,  $\text{wd} \geq 2.5$ ) and weakens toward phase boundaries. *Right:* majority phase per cell. Memorization occupies the high-lr, low-wd corner.

By weight decay, G<F increases monotonically from 72% at  $\text{wd}=1.43$  to 95% at  $\text{wd} \geq 2.93$ . F<G cases concentrate at the low-lr and high-lr boundaries (12 of 14 cases under the original detector-only ordering label, 11 of 13 under the canonical  $|\Delta\tau| \leq 500$  coincident threshold). This boundary clustering initially motivated the speed-dependent ordering hypothesis tested in Section 7; the targeted slow-grokking sweep there falsifies its strong form. We therefore treat these Stage 6 boundary cases as phase-margin irregularities (likely seed/hardware-level fluctuations and detector-resolution artefacts) rather than evidence for a systematic F<G regime. The interior of the Grokking phase is robustly G<F. Figure 8 shows the full spatial structure.

## 5.8 Decoupled Weight-Decay Ablation

*The two ablations in this section identify the conditions that produce the timing dissociation: E4 isolates the optimizer-side pathway that generates the lag and locates the regime boundary, and E5 varies generalization speed through the data path while holding that pathway fixed.*

The canonical sweep applies weight decay only to the decoder parameter group; embedding parameters are by default excluded. To distinguish whether the G<F ordering is driven by decoder weight decay, by embedding weight decay, or by their interaction, and to localise *when* the decay pressure matters, we run two coupled ablations at the canonical operating point  $(\text{lr}, \text{wd}^*) = (1.6 \times 10^{-3}, 2.5)$  with  $\tau_{\text{gen}}^{\text{ref}} = 17,000$  (source: `src/sweeps/e4_decoupled_wd.py`, results in `results/e4/results.csv`).

**2 × 2 ablation** —  $(\text{wd}_{\text{dec}}, \text{wd}_{\text{emb}}) \in \{0, \text{wd}^*\}^2$ . Three seeds per cell, 12 runs total. Table 6 reports the per-cell phase distribution and ordering. Three observations follow, each conditional on the canonical  $(\text{lr}, \text{wd}^*)$  operating point and 3 seeds per cell. **(i)  $\text{wd}_{\text{dec}}=0$  abolishes grokking under this canonical setting:** both cells with zero decoder decay memorise in all 3 seeds (max test accuracy  $\leq 0.10$ ). Decoder weight decay therefore appears *necessary* for grokking under this canonical operating point; we read this as a local causal statement, not a universal one. The  $(0, \text{wd}^*)$  cell is informative for a separate reason: despite direct weight decay on the embedding parameters, it produces  $\tau_F$  but no  $\tau_{\text{gen}}$  (F\_only phase). This provides direct evidence that embedding Fourier alignment is *not sufficient* for generalisation, and the headline G<F result rules out the converse reading as well: with  $\tau_F$  falling *after*  $\tau_{\text{gen}}$  in 212/236 Grokking runs, ring consolidation cannot be a temporal prerequisite for generalisation either. **(ii)  $\text{wd}_{\text{dec}}=\text{wd}^*$  alone reproduces canonical G<F:** the  $(\text{wd}^*, 0)$  cell groks in all 3 seeds with mean  $\Delta\tau=6,500$  steps, matching the Stage 2 G<F plateau. **(iii) Adding  $\text{wd}_{\text{emb}}=\text{wd}^*$  on top of  $\text{wd}_{\text{dec}}=\text{wd}^*$  destabilises the canonical dynamics:** 1/3 seeds memorises, and the 2/3 that grok do so with  $\tau_F \in \{8,000, 10,000\}$ , arriving *before*  $\tau_{\text{gen}} \in \{19,500, 16,000\}$

Table 6: E4a —  $2 \times 2$  weight-decay ablation (3 seeds per cell). Phase counts are over the 3 seeds. For Grokking seeds the range of  $(\tau_{\text{gen}}, \tau_F, \Delta\tau)$  is reported. Source: `results/e4/results.csv`.

| wd <sub>dec</sub> | wd <sub>emb</sub> | phase (3 seeds)     | $\tau_{\text{gen}}$ | $\tau_F$         | ordering |
|-------------------|-------------------|---------------------|---------------------|------------------|----------|
| 0                 | 0                 | Mem $\times 3$      | —                   | —                | none     |
| 0                 | wd*               | Mem $\times 3$      | —                   | [8,000, 28,500]  | F_only   |
| wd*               | 0                 | Grok $\times 3$     | [10,000, 13,500]    | [15,000, 23,500] | G<F      |
| wd*               | wd*               | Mem+Grok $\times 2$ | [16,000, 19,500]    | [8,000, 10,000]  | F<G      |

Table 7: E4b — Decoder-WD shutoff schedule (3 seeds each) at  $\tau_{\text{gen}}^{\text{ref}}=17,000$ . Source: `results/e4/results.csv`.

| shutoff step                                  | phase (3 seeds) | $\tau_{\text{gen}}$ range | $\tau_F$ range   | max test acc |
|-----------------------------------------------|-----------------|---------------------------|------------------|--------------|
| 8,500 ( $0.5\tau_{\text{gen}}^{\text{ref}}$ ) | Mem $\times 3$  | —                         | [21,500, 24,000] | 0.47–0.73    |
| 17,000 ( $\tau_{\text{gen}}^{\text{ref}}$ )   | Grok $\times 3$ | [10,000, 13,500]          | [16,000, 17,500] | 1.00         |
| 34,000 ( $2\tau_{\text{gen}}^{\text{ref}}$ )  | Grok $\times 3$ | [10,000, 13,500]          | [16,000, 29,500] | 1.00         |

— an F<G ordering, reproducibly. Direct embedding decay therefore accelerates Fourier consolidation enough to precede generalisation, giving a second reproducible recipe for F<G beyond the low-lr regime observed in Stage 3. Taken together, the  $2 \times 2$  results reframe our central claim: *decoder-only weight decay* is the operating point under which G<F is observed; “weight decay drives grokking” is more accurately stated as “*decoder weight decay drives grokking*, and combining it with embedding decay flips the ordering”.

**Decoder-WD shutoff schedule.** Starting from the canonical  $(\text{wd}_{\text{dec}}, \text{wd}_{\text{emb}})=(\text{wd}^*, 0)$  configuration, we set  $\text{wd}_{\text{dec}}=0$  at step  $s^* \in \{0.5, 1, 2\} \cdot \tau_{\text{gen}}^{\text{ref}}$  (3 seeds each, 9 runs). Results in Table 7 localise the causal window. Shutoff at  $0.5\tau_{\text{gen}}^{\text{ref}}=8,500$ , *before* the canonical  $\tau_{\text{gen}}$ , eliminates grokking in all 3 seeds: max test accuracy plateaus at 0.47–0.73 (still a partial progress signal, but never crossing the 0.9 threshold used for  $\tau_{\text{gen}}$ ). Shutoff at  $\tau_{\text{gen}}^{\text{ref}}$  and at  $2\tau_{\text{gen}}^{\text{ref}}$  both preserve canonical grokking, with G<F in 3/3 seeds at each shutoff time. Decoder weight decay therefore appears to act *locally causally* during the pre- $\tau_{\text{gen}}$  window: removing it within  $[0, \tau_{\text{gen}})$  abolishes grokking, while shutting it off after  $\tau_{\text{gen}}$  does not rescind generalisation. Prolonged decay past  $\tau_{\text{gen}}$  delays  $\tau_F$  (median  $\tau_F$  at shutoff  $2\tau_{\text{gen}}^{\text{ref}}$  is 29,000 vs. 17,000 at  $\tau_{\text{gen}}^{\text{ref}}$ ), consistent with decay continuing to suppress the magnitude signal that drives the BIC changepoint.

Together these two ablations identify decoder weight decay as the optimizer-side condition that puts the network into the grokking regime under the tested conditions: within the canonical operating point  $(\text{lr}, \text{wd}^*)=(1.6 \times 10^{-3}, 2.5)$  on  $(a+b) \bmod 53$  with the 2-layer transformer used throughout and  $n=3$  seeds per cell, removing decoder weight decay abolishes grokking, retaining only decoder weight decay (with  $\text{wd}_{\text{emb}}=0$ ) reproduces the canonical G<F regime, and shutting decoder decay off after  $\tau_{\text{gen}}$  does not rescind generalisation. These are causal statements about the tested grid: within it, decoder weight decay is what drives the network into the grokking regime, and the embedding is shaped indirectly through it. The decoupling also identifies the boundary of the regime. Adding embedding decay on top of decoder decay flips the ordering to F<G, which locates the G<F phenomenon specifically in the decoder-only decay regime used throughout Stages 2–6 and identifies the weight-decay pathway as the mechanism that sets the lag.

## 5.9 Data-Subsampling Speed Control

E4 provides local causal evidence that decoder weight decay is a locally required optimizer-side condition for grokking; E5 asks whether the G<F ordering is tied to the absolute value of  $\tau_{\text{gen}}$  (“how late generalisation occurs on the weight-decay clock”) or only to its sign relative to  $\tau_F$ . Holding the canonical operating point  $(\text{lr}, \text{wd}_{\text{dec}}, \text{wd}_{\text{emb}})=(1.6 \times 10^{-3}, 2.5, 0)$  fixed, we vary the training-set fraction  $\rho \in \{0.20, 0.30, 0.40, 0.50\}$  of all  $p^2$  pairs and run 3 seeds per cell (12 runs, 50,000 steps each; source: `src/sweeps/e5_data_subsample.py`, results in `results/e5/results.csv`). Varying  $\rho$  moves  $\tau_{\text{gen}}$  through the data-coverage bottleneck while holding the architecture, optimizer, learning rate, and weight-decay schedule fixed. This is the relevant

Table 8: E5 — Data-subsampling speed control (3 seeds per  $\rho$ , canonical (lr, wd\*, 0)). Medians are over seeds with the timescale defined; “–” indicates the timescale is not detected.  $P(G < F)$  is the fraction of the 3 seeds with  $G < F$  ordering; for  $\rho=0.20$  all three seeds are Memorization, so  $\tau_{\text{gen}}$  is undefined and the ratio is not applicable (denoted “N/A” in the table). Source: `results/e5/results.csv`.

| $\rho$ | phase (3 seeds)                   | med. $\tau_{\text{gen}}$ (steps) | med. $\tau_F$ (steps) | med. $\Delta\tau$ (steps) | $P(G < F)$ |
|--------|-----------------------------------|----------------------------------|-----------------------|---------------------------|------------|
| 0.20   | Mem $\times 3$                    | –                                | 21,750                | –                         | N/A        |
| 0.30   | Grok $\times 3$                   | 13,000                           | 17,500                | 5,000                     | 1.00       |
| 0.40   | Comp $\times 2$ , Grok $\times 1$ | 2,000                            | 20,000                | 18,000                    | 1.00       |
| 0.50   | Comp $\times 2$ , Grok $\times 1$ | 1,500                            | 12,000                | 10,500                    | 1.00       |

control here precisely because it operates on a different pathway from the weight-decay clock that shapes  $\tau_F$ : it changes how much data the model must generalise from, not how fast the regulariser reshapes the embedding, so any ordering change it produces would have to come from generalization timing itself.

Table 8 summarises the per- $\rho$  outcome. Three observations follow. **(i)** At  $\rho=0.20$  the model memorises in 3/3 seeds (max test accuracy  $\leq 0.02$ );  $\tau_F$  is still detected in 2/3 seeds, yielding an F\_only phase that replicates the (wd<sub>dec</sub>=0, wd<sub>emb</sub>=wd\*) cell of Table 6 — Fourier alignment is again confirmed insufficient for generalisation, here bottlenecked by data volume rather than by the decay target. **(ii)** At  $\rho=0.30$  the model enters the canonical grokking window (median  $\tau_{\text{gen}}=13,000$ , median  $\tau_F=17,500$ , median  $\Delta\tau=5,000$ ), overlapping the Stage 2 bootstrap interval (median  $\Delta\tau=7,750$ , 95% CI [6,000, 9,750]; see Table 13). **(iii)** At  $\rho \in \{0.40, 0.50\}$  the model exits the delayed-generalisation regime:  $\tau_{\text{gen}}$  collapses by roughly an order of magnitude to  $\{1,500, 2,000\}$  (Comprehension phase in 4/6 seeds), while  $\tau_F$  shifts less, and the gap  $\Delta\tau$  widens to  $\{10,500, 18,000\}$ .

The headline result is that  $\tau_{\text{gen}}$  varies by roughly  $10\times$  across the four  $\rho$  values while the ordering label is  $G < F$  in 9/9 runs (3 seeds each at  $\rho \in \{0.30, 0.40, 0.50\}$ ) where  $\tau_{\text{gen}}$  is defined (100%); the  $\rho=0.20$  row contributes no Grokking runs and is therefore not part of this denominator. This is the cleanest available dissociation of the  $G < F$  ordering from the absolute value of  $\tau_{\text{gen}}$  at the canonical operating point: neither a slower-than-canonical  $\tau_{\text{gen}}$  (as in  $\rho=0.30$ ) nor a faster-than-canonical  $\tau_{\text{gen}}$  (as in  $\rho \in \{0.40, 0.50\}$ ) moves the ordering off  $G < F$ , under a manipulation that leaves the weight-decay clock untouched.

E5 therefore supports only the *weak* form of the speed-dependent hypothesis of Section 7, namely that  $G < F$  is not an artefact of the canonical  $\tau_{\text{gen}}$  value. It does not test the *strong* form, which predicts an  $F < G$  transition once  $\tau_{\text{gen}}$  enters the slow-grokking margin above roughly 35,000 steps: all reachable E5 cells have  $\tau_{\text{gen}} \leq 13,000$  steps, an order of magnitude below that regime. The strong form requires a targeted slow-grokking test, which we perform in Section 7.

## 6 Discussion

### 6.1 Embedding Geometry as a Lagging Indicator

Our results are compatible with a picture of grokking that distinguishes *mechanism formation* from *embedding consolidation*. Nanda et al. (2023) showed that the Fourier computation circuit forms—and progress measures improve—before the generalization jump  $\tau_{\text{gen}}$ . Our complementary finding is that the token *embedding geometry* (measured by  $F_{\text{corr}}$ ) continues to consolidate for thousands of steps *after*  $\tau_{\text{gen}}$ .

Our results are *consistent with* a three-stage timing picture: (1) single-harmonic logit Fourier alignment onset  $\tau_{\text{circuit}}$ , a detector-specific proxy for circuit formation (**this work**, via  $F_L$ ); (2) generalization onset  $\tau_{\text{gen}}$ ; (3) embedding-geometry consolidation  $\tau_F$  (**this work**). All three events are measured in the same runs, without relying on Nanda et al.’s separate dataset. The strong-form claim “Fourier circuit forms first” is contingent on accepting single-harmonic logit projection as the operational proxy for circuit formation; under coarser loss-based proxies the  $C < G$  rate is substantially lower (Section 6.6, E3). The three intervals are approximately equal: median  $\tau_{\text{gen}} - \tau_{\text{circuit}} \approx \tau_F - \tau_{\text{gen}} \approx 6,000\text{--}6,250$  steps, so the total span  $\tau_F - \tau_{\text{circuit}} \approx 13,000$  steps is split nearly evenly between stages (1→2) and (2→3). The familiar Fourier ring visible in PCA visualizations

reflects stage (3), a post-generalization structural consequence, not stage (1), whose rise precedes the generalization jump.

A further finding is consistent with this view: F\_only runs (Fourier geometry detected but no generalization) exist at  $\text{lr} \approx 5.9 \times 10^{-3}$ , showing that Fourier geometric organization is *insufficient* for generalization near the phase boundary. Together,  $G < F$  and F\_only runs suggest that *this measured  $F_{\text{corr}}$ -based embedding alignment* is neither synchronized with generalization (it lags  $\tau_{\text{gen}}$  in the  $G < F$  regime) nor predictive of it (Fourier structure can accumulate without triggering generalization in the F\_only regime). What we measure here is a logit-level single-harmonic Fourier alignment proxy for circuit onset (Section 4, the  $F_L$  definition), not a full multi-harmonic mechanistic verification of the grokked circuit. Whether the embedding-side  $F_{\text{corr}}$  lag corresponds to a delay in the complete multi-harmonic implementation, or only in its single-harmonic projection, requires further investigation with mechanistic circuit probes such as activation patching or component-level Fourier ablations.

**Limits of the proxy: dissociation is between subspaces, not full mechanisms.** We use  $\tau_{\text{circuit}}$  as a logit-side Fourier proxy rather than as a claim that the complete circuit has been reverse-engineered at that step. The full modular-addition algorithm is implemented by a distributed transformer computation involving attention, MLP, and potentially multiple harmonics. Our claim is therefore narrower than “the algorithm forms before generalization”: the low-order answer-class *logit-Fourier subspace* identified by this proxy is already functionally necessary before the *embedding-Fourier ring* consolidates. The dissociation we establish is between an early load-bearing logit-Fourier subspace and a later visible embedding-Fourier geometry, not between a fully specified circuit and an irrelevant embedding matrix. Consistently, under coarser loss-based proxies the  $C < G$  rate falls to 63.6%/23.6% (Section 6.6, E3), so the proxy-level  $C < G < F$  headline is detector-dependent rather than universal. E7 itself is a 12-cell causal-localization probe (spanning seeds, weight decay, learning rate, task, and prime); the large- $N$  evidence for the embedding-ring lag rests on the 212/236 main-stage timing aggregate (236/260 including the post-hoc slow-grokking sweep).

**Division of labor between E7 and the timing aggregate.** E7 is a targeted causal-localization probe, not a high-powered phase-diagram estimate; its role is to test whether the timing proxy identifies a load-bearing subspace. The large- $N$  evidence for the embedding-ring lag itself comes from the 212/236 main-stage timing aggregate (236/260 with the post-hoc slow-grokking sweep) spanning multiple sweeps, two primes, and two tasks (Section 6.6). The two pieces of evidence answer different questions: timing characterises the regularity at population scale; E7 answers whether the regularity is causally meaningful at any single checkpoint.

**Practical implication.** Our dissociation suggests that embedding-based progress measures (PCA ring inspection, dictionary-learning probes) may systematically *underestimate* how far circuit formation has progressed during the memorization-to-generalization transition: in every fully grokked cell we tested, the logit-Fourier subspace was already causally load-bearing at checkpoints where the embedding ring had not yet consolidated. Practitioners who monitor training via embedding geometry should therefore treat a clean Fourier ring as a *lagging* indicator of algorithmic capability, not a concurrent one.

The three-stage decomposition we observe is also consistent with broader views of learning as a sequence of discrete events: Saxe et al. (2014) showed that deep linear networks learn singular modes one at a time, and Michaud et al. (2023) argue that neural scaling reflects the sequential acquisition of discrete “quanta” of skill.  $C < G < F$  is one concrete instantiation of such stage-wise dynamics in a setting where each stage has a measurable structural signature.

## 6.2 Connection to Rank, Flatness and Complexity Timing

Our  $\tau_F$  marks a discrete BIC-detectable event in a single-harmonic projection of the embedding matrix. A growing literature reports that grokking is accompanied by late-phase reductions in solution complexity along several distinct axes: effective rank of weight matrices and Hessian spectra collapsing toward a low-dimensional subspace (Mohamadi et al., 2024), and progress measures based on loss-landscape and feature-statistic transitions (Miller et al., 2024). Our  $F_{\text{corr}}$  trajectory is consistent with this family of



observations because the Fourier subspace alignment we measure is precisely a low-rank reorganisation of the embedding row space onto a 2-dimensional cyclic-group invariant subspace, so a sharp rise in  $F_{\text{corr}}$  implies an accompanying drop in the effective rank of the embedding restricted to its task-relevant subspace. Two predictions follow. First, in larger transformers solving cyclic-group tasks, the BIC changepoint of  $F_{\text{corr}}$  should approximately co-locate with the step at which the embedding matrix’s stable rank or participation ratio drops, providing a falsifiable bridge between our single-harmonic onset and rank-based complexity measures. Second, on tasks where the late-phase rank collapse is driven by non-Fourier structure (for example, the Toy Models of Superposition basis-rotation phenomena (Elhage et al., 2022)) we would expect  $\tau_F$  to lag or fail to detect, while a rank-based progress measure would still register the transition; this provides an experimentally clean separation between cyclic-group-specific and rank-generic notions of “circuit consolidation”.

### 6.3 Why the Metric Matters

We use null-corrected  $F_{\text{corr}}$  rather than raw  $F_{\text{raw}}$  because random embeddings already exhibit non-trivial  $F_{\text{raw}}$  in high dimensions ( $p = 53$ ,  $d = 256$ ): any projection of 53 points in 256 dimensions partially aligns with some harmonic by chance. The null-corrected  $F_{\text{corr}}$  removes this baseline and marks the step at which embeddings show *significantly above-chance* harmonic structure.

To assess whether null correction changes the conclusions of this paper, we applied the BIC changepoint estimator independently to  $F_{\text{corr}}$  and  $F_{\text{raw}}$  for all 8 sensitivity cells (Table 9). For cells with  $|\Delta\tau| \geq 3,000$  steps, the two metrics agree: 6 of 6 G<F cells remain G<F under raw Fourier, because the sharp post- $\tau_{\text{gen}}$  rise in  $F_{\text{raw}}$  dominates the changepoint detection regardless of baseline. The null correction matters most for marginal cases: the coincident cell (wd= 3.11) and the F<G outlier remain unchanged. We conclude that null correction does not alter the main finding for runs with substantial lags, but is conceptually important for calibrating the detection threshold and avoiding false positives in the coincident regime.

Table 9: Ordering under null-corrected vs. raw Fourier for 8 sensitivity cells (canonical detector config). For cells with  $|\Delta\tau| \geq 3,000$  both metrics agree; null correction is relevant primarily near the coincident boundary.

| Cell           | Category   | Order ( $F_{\text{corr}}$ ) | Order (raw) | Flipped?   |
|----------------|------------|-----------------------------|-------------|------------|
| wd=1.71, s7    | strong G<F | G<F                         | G<F         | No         |
| wd=2.76, s42   | strong G<F | G<F                         | G<F         | No         |
| wd=1.52, s42   | medium G<F | G<F                         | G<F         | No         |
| wd=2.18, s42   | medium G<F | G<F                         | G<F         | No         |
| wd=2.45, s7    | weak G<F   | G<F                         | G<F         | No         |
| wd=3.50, s7    | weak G<F   | G<F                         | G<F         | No         |
| wd=3.11, s2025 | coincident | coincident                  | coincident  | No         |
| lr=6.8e-4, s7  | F<G        | F<G                         | F<G         | No         |
| <b>Total</b>   |            | 6/8 G<F                     | 6/8 G<F     | <b>0/8</b> |

### 6.4 Connection to Superposition and Polysemanticity

The C<G<F ordering carries a tentative implication for representation analysis in larger models. Stage C ( $\tau_{\text{circuit}}$ ) marks the BIC-detected onset of single-harmonic Fourier alignment in the logit space, our proxy for circuit formation; Stage F ( $\tau_F$ ) marks the later moment at which the token embedding itself collapses onto the cyclic-group representation that this proxy already detects in the logits. In our 2-layer transformer the two events are separated by 5,000–10,000 steps. The natural reading is that, during the C<G window, the network already implements the correct algorithm but does so on representations that are still polysemantic: the embeddings have not yet specialised onto the cyclic-group basis, so the circuit must operate through a superposition of features rather than through a clean one-feature-per-direction code. This is consistent with the broader picture of superposition in transformers (Elhage et al., 2022) and with the observation that monosemantic features can be recovered from polysemantic activations only with explicit disentangling (Bricken et al., 2023; Huben et al., 2024). Whether an analogue of  $\tau_F$ -style geometry consolidation appears in larger models is an empirical question we leave open. The consequence that follows directly from our measurements

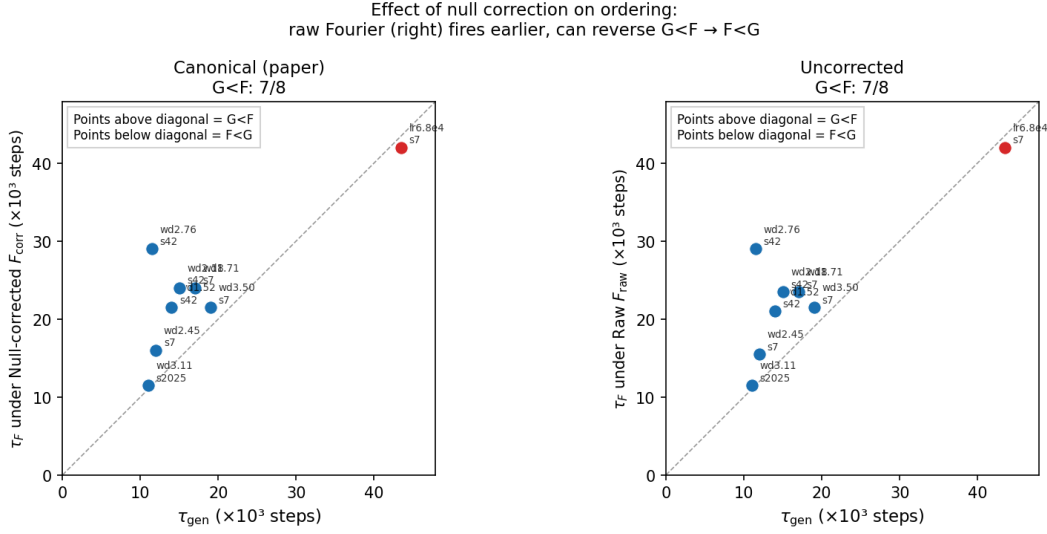


Figure 9: Scatter of  $\tau_F$  under null-corrected (left) vs. raw (right) Fourier for 8 sensitivity cells. Points above the diagonal indicate  $G < F$  ordering; points below indicate  $F < G$ . Blue points remain  $G < F$  under both metrics; the red point ( $F < G$  outlier) remains below the diagonal. No ordering flips across the two metrics.

is internal: on this same model, between  $\tau_{\text{circuit}}$  and  $\tau_F$  the circuit operates on a representation that the embedding-side metric  $F_{\text{corr}}$  has not yet identified as cyclic-group-aligned, so methods that rely on the embedding’s geometric handle (linear probes, dictionary learning) may underdetect the circuit during that window even though the logit-level Fourier proxy already flags it.

## 6.5 Detector Robustness

A natural concern is whether the  $G < F$  finding depends on the specific choice of detector hyperparameters: the null-update interval, the EMA smoothing coefficient, and the BIC slope threshold. The null-update interval axis doubles as the artefact check for the LOCF (piecewise-constant) null schedule introduced in Section 4: a factor-of-10 change in the refresh cadence bounds the maximum  $\tau_F$  displacement that the LOCF discontinuities could induce. To address this, 8 cells are selected from the Stage 2+3 classification table by a fixed stratification rule—cells stratified across the measured  $|\Delta\tau|$  distribution (large, medium, small), plus the coincident cell and the original Stage 3  $F < G$  outlier (detector-stable but not replicated under the cross-hardware re-run of Section 7)—and all  $3 \times 4 \times 3 = 36$  detector configurations are applied post-hoc to the same saved trajectories. Selecting the subset by a stratification rule applied to the already-classified table, rather than picking individual cells after inspecting their detector behaviour, reduces the risk of cherry-picked robustness.

We define the **Robustness Score**  $RS = N_{G < F} / N_{\text{total}}$  for each cell, where  $N_{\text{total}} = 36$ . Results are shown in Figure 10. The five cells with  $|\Delta\tau| \geq 5,000$  steps achieve  $RS = 1.00$  (36/36 configurations yield  $G < F$ ). The two cells with  $|\Delta\tau| \approx 2,000$ –3,000 steps achieve  $RS = 0.83$  and  $RS = 0.75$ , with the remaining configurations classifying as *coincident* rather than  $F < G$ . Crucially, **no  $G < F$  cell is ever classified as  $F < G$**  under any configuration. The mean  $RS$  across all seven non- $F < G$  cells is **0.94**, confirming that the  $G < F$  finding forms a robustness plateau over the detector parameter space.

**The two  $F < G$  cases.** Both  $F < G$  runs are real within their original trajectories, and each has an identifiable origin at the edge of the Grokking phase.

$lr = 6.8 \times 10^{-4}$ , *seed* 7 ( $\Delta\tau = -1,500$ ,  $RS = 0/36$ ). The label is detector-stable: all 36 configurations return  $F < G$ . Trajectory inspection locates its origin.  $F_{\text{corr}}$  at  $\tau_{\text{gen}}$  is 0.015, at the detection floor, while  $\tau_{\text{gen}} = 43,500$  is the latest generalization time in our dataset. At this very low learning rate, the weight-decay-to-learning-rate

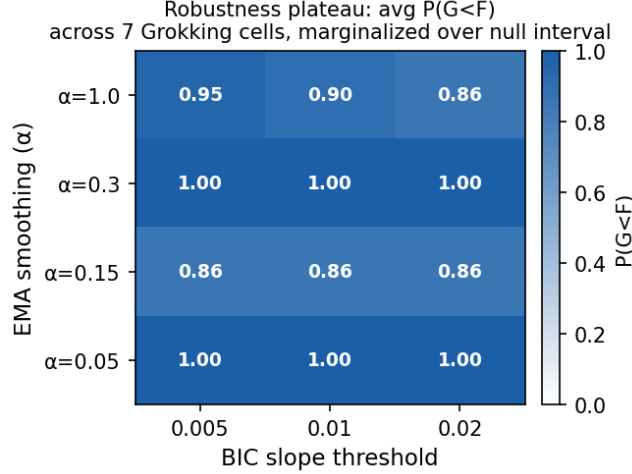


Figure 10: **Robustness plateau.** Each cell shows  $P(G < F)$  for a given (BIC slope threshold, EMA  $\alpha$ ) combination, averaged over null-update intervals  $\{500, 2500, 5000\}$  and over all seven Grokking cells. Strong and medium  $G < F$  cells achieve 1.00 across all configurations; weaker cells ( $|\Delta\tau| \lesssim 3,000$  steps) degrade gracefully to *coincident*, never to  $F < G$ .

ratio is  $wd/lr \approx 3,700$ , compared to  $\approx 1,070$  for a typical  $G < F$  cell. The interpretation this supports is that strong relative weight decay slowly compresses the embeddings toward Fourier-structured geometry during the long wait before generalization, accumulating a barely detectable  $F_{\text{corr}}$  signal before the sharp  $\tau_{\text{gen}}$  transition, which is the same weight-decay pathway that produces the lag elsewhere, running at an extreme parameter ratio. The re-run of this cell on different hardware in the slow-grokking sweep of Section 7 returns  $\Delta\tau = +9,000$  ( $G < F$ ), placing it within the ordinary seed-level spread of the main population.

$lr = 4.32 \times 10^{-3}$ , *seed 42* ( $\Delta\tau = -54,500$ , RS not assessed). This is the most extreme case:  $\tau_F = 13,000$  but  $\tau_{\text{gen}} = 67,500$  (extended to 80k steps), with  $\text{max\_test\_acc} = 0.965$  (below 1.0, indicating incomplete generalization). This run sits on the Grokking/Memorization phase boundary and never completes generalization, so it falls outside the population the  $G < F$  claim is about: at high  $lr$  the model develops embedding Fourier structure early while the incomplete generalization process unfolds on a much longer timescale.

Both cases sit at the edges of the Grokking phase rather than in its interior, and neither survives as a counterexample: the low- $lr$  case replicates as  $G < F$  when re-trained (Section 7), and the high- $lr$  case does not fully grok. The targeted slow-grokking sweep of Section 7 tests the phase margin directly, with 24/24 purpose-built cells landing  $G < F$ .

## 6.6 Post-hoc Robustness

The Detector-Robustness analysis above varies the changepoint pipeline while holding the metric fixed. This section varies the *metric* while holding the pipeline fixed, using dense embedding trajectories re-saved from the same 8 sensitivity cells (same ( $lr$ ,  $wd$ ,  $seed$ ) combinations and seeds as Section 6.5; embedding matrices recorded every 500 steps, 50,000 steps per run). All eight analyses below are post-hoc and add negligible compute beyond loading the `.npz` trajectory files and the per-run result CSVs (`results/posthoc/`).

**Null threshold and permutation count (`q2_null_sensitivity`).** We vary the two free parameters of the permutation null: the percentile  $p \in \{90, 95, 99\}$  and the number of permutations  $n \in \{50, 100, 200\}$  (9 configurations total). The canonical detector is ( $p=95, n=100$ ). For each configuration we recompute  $F_{\text{null},p}$  at every log step from the saved embeddings, re-derive  $F_{\text{corr}}$ , and re-run the EMA+BIC changepoint detector. Table 10 reports the per-cell  $\Delta\tau_F$  relative to canonical. Across all  $8 \times 9 = 72$  reconfigurations,  $|\Delta\tau_F| \leq 500$  steps (one log-bin), and for five of the eight cells every one of the nine configurations returns an identical  $\tau_F$

Table 10: Null-correction sensitivity. Per-cell  $\Delta\tau_F$  (in training steps) across the 9 null configurations relative to canonical ( $p_{95}, n=100$ ). When the three  $n$ -values within a percentile column disagree we report the full range; “0” means all three agree with canonical. Source: `results/posthoc/q2_null_sensitivity.csv`.

| cell (category)           | p90  | p95     | p99  |
|---------------------------|------|---------|------|
| wd1.71_s7 (strong G<F)    | −500 | 0       | 0    |
| wd2.76_s42 (strong G<F)   | 0    | 0       | 0    |
| wd1.52_s42 (medium G<F)   | 0    | 0, +500 | +500 |
| wd2.18_s42 (medium G<F)   | −500 | −500, 0 | 0    |
| wd2.45_s7 (weak G<F)      | 0    | 0       | 0    |
| wd3.50_s7 (weak G<F)      | 0    | 0       | 0    |
| wd3.11_s2025 (coincident) | 0    | 0       | 0    |
| lr6.8e4_s7 (F<G)          | 0    | 0       | 0    |

Table 11: Multi-harmonic  $\tau_F^K$  and ordering at  $K \in \{1, 2, 3, 5\}$ . Every cell preserves its canonical ordering label at all  $K$ ;  $\tau_F^K$  shifts are bounded by  $[-1,500, +6,500]$  steps. Source: `results/posthoc/q3_multiharmonic.csv`.

| cell             | $\tau_{\text{gen}}$ | $K=1$  | $K=2$  | $K=3$  | $K=5$  |
|------------------|---------------------|--------|--------|--------|--------|
| wd1.71_s7        | 17,000              | 26,000 | 25,500 | 27,500 | 28,500 |
| wd2.76_s42       | 11,500              | 30,000 | 30,000 | 30,000 | 30,500 |
| wd1.52_s42       | 14,000              | 23,000 | 28,000 | 29,000 | 29,500 |
| wd2.18_s42       | 15,000              | 26,500 | 27,500 | 28,500 | 29,500 |
| wd2.45_s7        | 12,000              | 18,500 | 17,000 | 16,500 | 18,500 |
| wd3.50_s7        | 19,000              | 23,000 | 23,000 | 23,500 | 24,500 |
| wd3.11_s2025     | 11,000              | 14,000 | 15,500 | 16,500 | 17,500 |
| lr6.8e4_s7 (F<G) | 43,500              | 42,000 | 42,000 | 42,000 | 42,000 |

( $\Delta\tau_F = 0$ ). No cell changes its ordering label under any choice of  $(p, n)$ . This rules out the possibility that the reported G<F / F<G labels are artefacts of the canonical ( $p_{95}, n=100$ ) null.

**Multi-harmonic Fourier variant (q3\_multiharmonic).** Our canonical  $F_{\text{raw}}$  uses  $\max_k R^2(E_t \mid \text{span } Q_k)$ , which restricts attention to a single harmonic at each step. A natural generalisation projects onto the top- $K$  harmonics by per-step energy, yielding  $F_K(t)$  with  $K \in \{1, 2, 3, 5\}$  ( $K=1$  reproduces the canonical metric as a sanity check). We recompute the permutation null at matching  $K$  (100 shuffles), re-derive  $F_K^{\text{corr}}$ , and re-run the changepoint detector. Table 11 reports  $\tau_F^K$  and the resulting ordering label per cell. Every G<F cell retains G<F at all four  $K$  values (28/28), and the single F<G cell retains F<G at all four (4/4): the ordering is invariant to the harmonic-subspace width. Contrary to a natural worry, richer subspaces tend to detect  $\tau_F^K$  *later*, not earlier, because the top- $K$  subspace saturates more slowly and the BIC detector demands a correspondingly larger slope change (seven of eight cells shift by 0 to +6,500 steps between  $K=1$  and  $K=5$ , and none shift in the opposite direction beyond 500). The single-harmonic metric is therefore, if anything, optimistic about  $\tau_F$ : widening the subspace only enlarges  $\Delta\tau$  for G<F cells.

**Embedding-layer geometry trajectories (q8\_embedding\_geometry).** To test whether  $\tau_F$  coincides with a broader geometric consolidation, we compute four auxiliary metrics at every checkpoint: Frobenius norm  $\|E_t\|_F$ , participation ratio  $\text{PR} = (\sum \sigma_i^2)^2 / \sum \sigma_i^4$ , effective rank  $e^S$  with  $S$  the Shannon entropy of the normalised squared-SV spectrum, and mean cosine anisotropy  $A = \text{mean}_{i<j} |\cos(E_i, E_j)|$ . For each metric we detect a geometry changepoint  $\tau_{\text{geom}}$  via the same EMA+BIC pipeline and compare it run-by-run to  $\tau_F$ . Table 12 reports the per-cell alignment and Figure 11 overlays the four trajectories with  $\tau_{\text{gen}}$ ,  $\tau_F$ , and  $\tau_{\text{geom}}$  markers on a log-step axis.

Every one of the  $8 \times 4 = 32$  (cell, metric) pairs gives  $\tau_{\text{geom}} < \tau_F$ , most by a substantial margin. A consistent temporal ordering emerges within each cell:  $\tau_{\|E\|_F} \lesssim \tau_{\text{PR}} \lesssim \tau_{e^S} \lesssim \tau_A < \tau_F$ . Across cells, median gaps are  $\tau_F - \tau_{\|E\|_F} \approx 15,000$ ,  $\tau_F - \tau_{e^S} \approx 13,000$ ,  $\tau_F - \tau_A \approx 6,500$  steps; anisotropy is the geometry metric closest to

Table 12: Geometry changepoint  $\tau_{\text{geom}}$  for four embedding-level metrics, compared per cell against canonical  $\tau_F$ . Every  $(\text{cell}, \text{metric})$  pair gives  $\tau_{\text{geom}} < \tau_F$ . Source: `results/posthoc/q8_geometry.csv`.

| cell             | $\tau_{\text{gen}}$ | $\tau_F$ | $\ E\ _F$ | PR     | $e^S$  | A      |
|------------------|---------------------|----------|-----------|--------|--------|--------|
| wd1.71_s7        | 17,000              | 25,500   | 10,500    | 14,000 | 15,000 | 22,000 |
| wd2.76_s42       | 11,500              | 30,000   | 8,000     | 9,000  | 9,500  | 15,500 |
| wd1.52_s42       | 14,000              | 23,500   | 11,000    | 13,000 | 14,500 | 17,000 |
| wd2.18_s42       | 15,000              | 26,500   | 9,000     | 10,000 | 11,000 | 15,500 |
| wd2.45_s7        | 12,000              | 18,500   | 8,000     | 9,500  | 10,500 | 14,000 |
| wd3.50_s7        | 19,000              | 23,000   | 8,000     | 8,000  | 8,000  | 8,000  |
| wd3.11_s2025     | 11,000              | 14,000   | 8,000     | 8,000  | 8,500  | 12,500 |
| lr6.8e4_s7 (F<G) | 43,500              | 42,000   | 8,000     | 9,500  | 10,500 | 12,500 |

Table 13: E6 — Non-parametric bootstrap 95% CIs ( $n_{\text{boot}}=1000$ ).  $n_{\Delta\tau}$  is the number of rows with both  $\tau_{\text{gen}}$  and  $\tau_F$  detected. Median  $\Delta\tau$  and  $P(G<F)$  are reported as point estimate with 95% percentile interval. The row-bs column resamples per-run rows as independent units; the cluster-bs column resamples (lr, wd) cells with replacement, keeping all seeds inside a cell together, and is the seed-clustering-aware interval (Table 14 gives the full comparison including Stages 5A/5B). The Stage 6 cluster estimate is computed over all 147 rows rather than the 126 detected pairs, hence the 0.75 vs 0.76 point difference; the cluster bootstrap was not run for E4. Sources: `results/posthoc/e6_bootstrap_ci.csv`, `results/posthoc/cluster_bootstrap_ci.csv`.

| source            | $n$ | $n_{\Delta\tau}$ | med. $\Delta\tau$ (steps) | $P(G<F)$ row-bs   | $P(G<F)$ cluster-bs |
|-------------------|-----|------------------|---------------------------|-------------------|---------------------|
| Stage 2 (wd)      | 30  | 30               | 7,750 [6,000, 9,750]      | 0.97 [0.90, 1.00] | 0.97 [0.90, 1.00]   |
| Stage 3 (lr)      | 30  | 25               | 4,000 [2,500, 9,500]      | 0.77 [0.60, 0.90] | 0.77 [0.57, 0.97]   |
| Stage 6 (2D grid) | 147 | 126              | 6,000 [5,000, 6,500]      | 0.76 [0.70, 0.83] | 0.75 [0.66, 0.84]   |
| Stage 4 (circuit) | 60  | 54               | 6,000 [5,000, 8,500]      | 0.87 [0.78, 0.95] | 0.87 [0.73, 0.97]   |
| E4 (ablation)     | 21  | 11               | 4,500 [2,500, 10,000]     | 0.43 [0.24, 0.67] | —                   |

$\tau_F$ , but still precedes it by more than a log-bin in every cell (minimum gap 1,500 steps). This falsifies the hypothesis that  $\tau_F$  is a late readout of generic embedding-norm or rank growth: Fourier alignment is a distinct, later event that is *selected from* an already consolidated geometric substrate. The same pattern holds for the F<G cell (lr6.8e4\_s7): all four  $\tau_{\text{geom}}$  values precede  $\tau_F$  by 29,500–34,000 steps and precede  $\tau_{\text{gen}}$  by 31,000–35,500 steps, showing that the geometry-first ordering is not specific to the G<F regime.

**E6 — Bootstrap confidence intervals (e6\_bootstrap\_ci).** To quantify the sampling uncertainty in the headline summaries used throughout the paper, we run a non-parametric percentile bootstrap ( $n_{\text{boot}}=1000$ ) over the per-run rows of the five main result CSVs: Stage 2 (wd-sweep at lr= $1.6 \times 10^{-3}$ ), Stage 3 (lr-sweep at wd=2.5), Stage 6 (2D (lr, wd) grid), Stage 4 (three-way ordering with  $\tau_{\text{circuit}}$ ), and E4 (decoupled weight-decay ablation).  $\Delta\tau$  is computed only where both changepoints are detected;  $P(G<F)$  is computed over all rows in each source, so a source containing many memorisation or F\_only cells naturally reports a lower proportion. Table 13 reports the 95% CIs, with the cell-level cluster-bootstrap interval (which keeps all seeds of a hyperparameter cell together) alongside the row-level interval. Across the four sweep-type sources the median  $\Delta\tau$  is strictly positive at the 95% level (lower bound  $\geq 2,500$  steps in every case), and  $P(G<F)$  is bounded away from 0.5 in each sweep, with the highest- $n$  source (Stage 6,  $n=147$ ) giving the tightest interval [0.70, 0.83]. E4’s lower  $P(G<F)=0.43$  [0.24, 0.67] reflects the by-design presence of non-grokking ablation cells (the (0, 0) and (0, wd\*) rows of Table 6); it is not evidence against G<F in the grokking regime.

**E3 — Comparison with Nanda et al. progress measures (e3\_nanda\_analysis).** A natural concern is that our  $F_L$ -based  $\tau_{\text{circuit}}$  might simply rediscover, in a more elaborate form, what the canonical restricted-logit and excluded-loss progress measures of Nanda et al. (2023) already detect. We therefore recompute both Nanda metrics on the trajectories of the 60 Stage 2+3 cells (the same cells used to measure  $\tau_{\text{circuit}}$  in Section 5.4; here the classifier is rerun independently on each trajectory and labels 55 cells Grokking and 5 Memorization, off-by-one from the 54 Grokking runs reported in Section 5.4 because one borderline

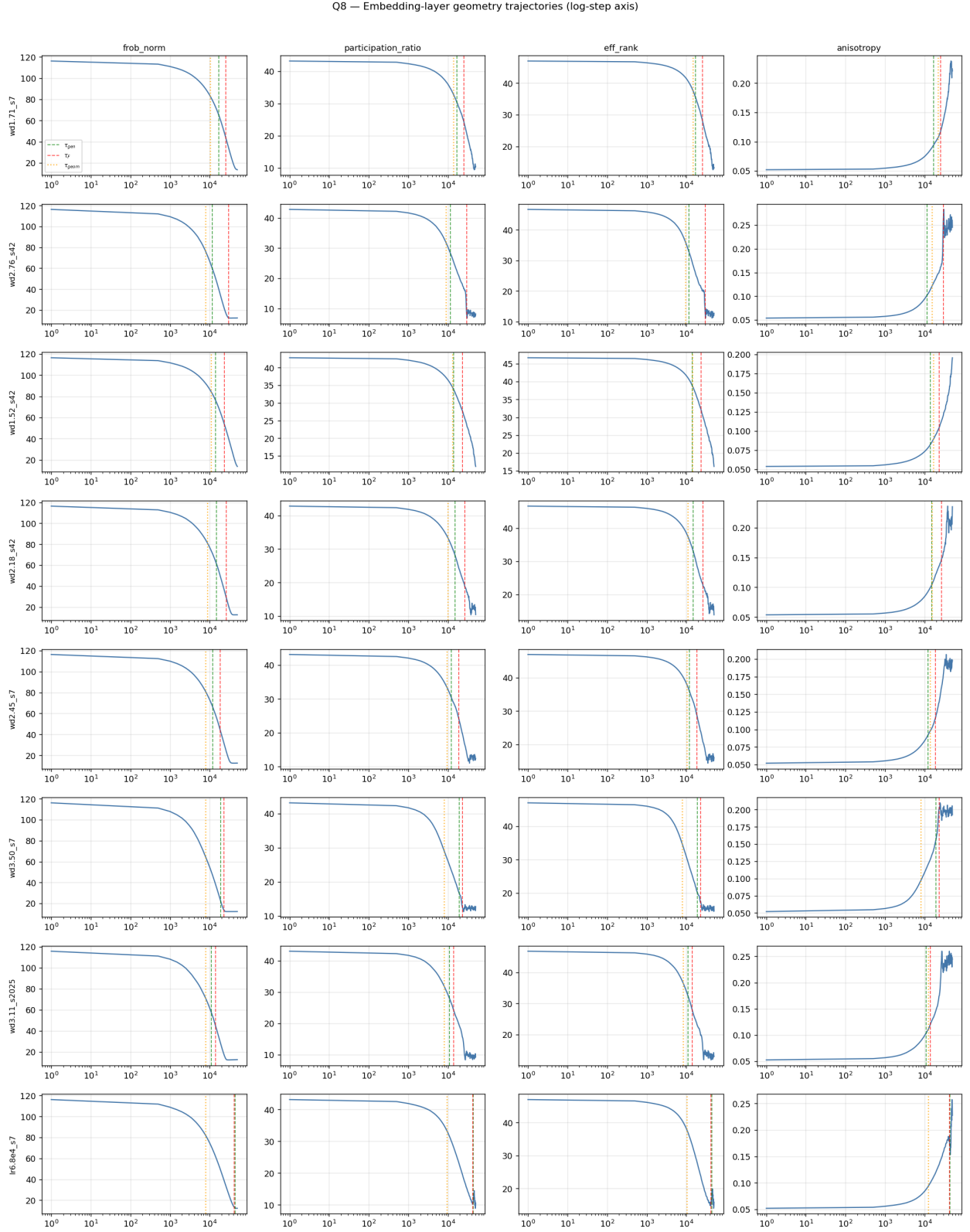


Figure 11: Embedding geometry trajectories. Four geometry metrics (columns:  $\|E\|_F$ , PR,  $e^S$ , anisotropy) for each of the 8 sensitivity cells (rows) on a log-step axis. Green dashed =  $\tau_{\text{gen}}$ , red dashed =  $\tau_F$ , orange dotted =  $\tau_{\text{geom}}$  for the metric shown. Every panel has  $\tau_{\text{geom}}$  strictly left of  $\tau_F$ .



Table 14: E6 supplementary — Per-source comparison of row-level bootstrap (Table 13) against a cell-level cluster bootstrap that resamples (lr, wd) cells with replacement and keeps all seeds inside a cell together ( $n_{\text{boot}}=1000$  each). Cluster intervals are 33–40% wider on the three sources where seed clustering is non-degenerate (Stage 3, Stage 6, Stage 4 retraining); point estimates shift by  $< 1\%$ . Stage 2 / Stage 5A / Stage 5B sit at  $G<F$  rates of 97% / 100% / 100% respectively, so intra-cluster variance is mechanically near zero and the two bootstraps coincide. Source: `results/posthoc/cluster_bootstrap_ci.csv`. The Stage 6 row-bootstrap point estimate differs slightly from Table 13 (0.75 vs 0.76) because this table bootstraps all  $n=147$  rows, while Table 13 conditions on the  $n_{\Delta\tau}=126$  rows where both onsets are detected.

| source              | $n_{\text{rows}}$ | $n_{\text{cells}}$ | P( $G<F$ ) row-bs | P( $G<F$ ) cluster-bs |
|---------------------|-------------------|--------------------|-------------------|-----------------------|
| Stage 2 (wd)        | 30                | 10                 | 0.97 [0.90, 1.00] | 0.97 [0.90, 1.00]     |
| Stage 3 (lr)        | 30                | 10                 | 0.77 [0.60, 0.90] | 0.77 [0.57, 0.97]     |
| Stage 5A (mul)      | 30                | 10                 | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00]     |
| Stage 5B ( $p=97$ ) | 30                | 10                 | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00]     |
| Stage 6 (2D grid)   | 147               | 49                 | 0.75 [0.69, 0.82] | 0.75 [0.66, 0.84]     |
| Stage 4 (circuit)   | 60                | 20                 | 0.87 [0.78, 0.95] | 0.87 [0.73, 0.97]     |

cell crosses the sustained-accuracy threshold under the E3 reclassification but not in the original Stage 4 pass) and rerun the *same* EMA-smoothed BIC changepoint pipeline ( $\alpha=0.15$ ) on each of the three scalar series. Two findings emerge (Table 15). First, the *relative* cell-by-cell ordering of the three detectors is moderately rank-consistent: Spearman  $\rho=0.54$  ( $p=7.7\times 10^{-6}$ ) for excluded loss and  $\rho=0.42$  ( $p=8.1\times 10^{-4}$ ) for restricted loss, both versus our  $\tau_{\text{circuit}}$  over all 60 paired runs. The three detectors are therefore *related but not interchangeable* scalar substitutes for a single underlying event: the moderate rank agreement is consistent with all three tracking some aspect of circuit formation, but the rank correlation is far from unity, so the single-harmonic  $F_L$  proxy is not a drop-in substitute for a fully multi-harmonic mechanistic circuit signal. Second, the *absolute* timing differs systematically: the median Nanda changepoint lands 2,500 steps later (excluded) and 4,750 steps later (restricted) than ours, and agreement to within  $\pm 2,000$  steps holds in only 38% and 33% of pairs respectively. The downstream consequence is the  $C<G$  rate among the 55 Grokking-phase runs: our  $F_L$ -based  $\tau_{\text{circuit}}$  reports strict  $C<G$  in 48/55 (87.3%) of cells, the excluded-loss changepoint in 38/55 (69.1%), and the restricted-loss changepoint in only 16/55 (29.1%). The full three-stage  $C<G<F$  rate falls correspondingly from 44/55 (80.0%, ours) to 35/55 (63.6%, excluded) and 13/55 (23.6%, restricted).

**Reconciling Stage 4’s 88.9% with E3’s 80.0% (“ours” detector).** Stage 4 reports 50/54=92.6%  $C\leq G$  and 48/54=88.9%  $C<G<F$ , while the E3 “ours” detector on the same nominal 60 (lr, wd, seed) cells gives 48/55=87.3%  $C<G$  and 44/55=80.0%  $C<G<F$ . The gap is too large for the one-cell phase reclassification (54 vs 55 Grokking). What actually differs is that Stage 4 and E3 are *independent training runs* of the same training script with the same nominal seeds, executed on different GPU/CUDA configurations (Stage 4 on a local RTX 4080, E3 on a rented A800). Across the 54 cells classified as Grokking under both runs, the per-cell  $\tau$  values differ: median  $|\Delta\tau_{\text{gen}}| = 2,000$  steps (90th percentile 7,200, max 30,500), median  $|\Delta\tau_F| = 3,250$ , and the three-way ordering label reclassifies on 8/60 cells. We attribute this to non-determinism that survives `torch.backends.cudnn.deterministic=True` when sweeps are run on different GPUs and CUDA versions. The detector itself ( $F_L$  + EMA + 1-break BIC) is identical between Stage 4 and E3; the 88.9% vs 80.0% gap is therefore a measurement of seed-cluster variance under cross-hardware non-determinism, not a detector disagreement. We therefore report Stage 4’s 48/54 result as the primary documented run (it is the configuration described throughout Sections 4–5.4) and E3’s 44/55 result as an independent replication estimate, treating the  $C<G<F$  rate as a range [80.0%, 88.9%] across two realizations rather than a single point estimate.

Under a standard EMA+BIC detector, then, the loss-based progress measures do not yield a clean  $C<G$  ordering. Restricted loss inverts the conclusion in three quarters of grokking cells. We read this not as a contradiction of Nanda et al., whose visual analysis already shows early restricted-loss decline, but as evidence that scalar loss-based progress measures are *too smooth* for changepoint quantification: the per-step

Table 15: E3 — Run-by-run comparison of  $\tau_{\text{circuit}}$  detectors on the 60 Stage 4 runs (55 Grokking, 5 Memorization). “Median  $\Delta$ ” is the per-pair difference (Nanda – ours) in steps;  $|\Delta| \leq 2,000$  is the fraction of pairs that agree to within  $\pm 2,000$  steps; Spearman  $\rho$  is computed across all 60 pairs. The C<G and C<G<F columns count the 55 Grokking-phase runs only. Source: `runs/e3_nanda/e3_agreement_stats.csv` and `results_e3_nanda.csv`.

| detector              | median $\Delta$ (steps) | $ \Delta  \leq 2,000$ | Spearman $\rho$                 | P(C<G)        | P(C<G<F)      |
|-----------------------|-------------------------|-----------------------|---------------------------------|---------------|---------------|
| ours ( $F_L$ )        | —                       | —                     | —                               | 0.873 (48/55) | 0.800 (44/55) |
| Nanda excluded loss   | +2,500                  | 0.38                  | 0.54 ( $p=7.7 \times 10^{-6}$ ) | 0.691 (38/55) | 0.636 (35/55) |
| Nanda restricted loss | +4,750                  | 0.33                  | 0.42 ( $p=8.1 \times 10^{-4}$ ) | 0.291 (16/55) | 0.236 (13/55) |

dynamic-top- $K$  Fourier projection used by  $F_L$  resolves circuit formation sharply enough that the C<G boundary becomes detector-quantifiable rather than merely eye-visible.

**Detector-specification sensitivity (w1\_multibreak\_check, w5\_taugen\_threshold).** We add two checks targeting the two specification choices most likely to be challenged: the 1-break formulation of  $\tau_F$  and the 0.90 accuracy threshold of  $\tau_{\text{gen}}$ . **Multi-break check.** On the same 8 representative cells used elsewhere in Section 6.6, we refit  $F_{\text{corr}}(\log t)$  with 0, 1, and 2 breaks and compare BIC. A 2-break model is BIC-preferred over a 1-break model in 7/8 cells, but the gain comes from adding a *late* second knee that captures the post-onset saturation plateau (the second break  $\text{bp}_2$  falls in 26,000–42,000 steps, well after  $\tau_{\text{gen}}$ ); the *early* break  $\text{bp}_1$  of the 2-break fit lands within median  $|\Delta|=1,500$  steps (maximum 4,000) of the canonical 1-break  $\tau_F$ . Read as a sanity check rather than a contradiction: the 1-break formulation is not hiding a secondary changepoint that would shift  $\tau_F$  outside the regime where  $G<F$  holds; it is collapsing a smooth two-knee curve into the earlier of its two knees, which is the timing event of interest. **Multi-harmonic and cyclic-group irrep alternatives.** For  $\mathbb{Z}/p$  the non-trivial irreducible representations are 1-dimensional characters whose real-form 2D subspaces coincide with the Fourier planes used by  $F_{\text{raw}}$ , so an unrestricted mean over all  $(p-1)/2$  irreps is mathematically degenerate (by Parseval, after centering, it equals the constant  $2/(p-1)$  at every step). The closest non-degenerate alternative is to score the sum of the top- $m$  most-aligned harmonic  $R^2$  values, recovering  $\tau_F^{(m)}$  under the same EMA+BIC pipeline. We recomputed  $\tau_F^{(m)}$  for  $m \in \{1, 2, 3, 5\}$  on the saved embedding trajectories of the 8 sensitivity cells (`src/analysis/multi_harmonic_recompute.py`).  $\tau_F^{(m)}$  is consistently equal to or later than the canonical  $\tau_F^{(1)}$ , with median  $|\tau_F^{(5)} - \tau_F^{(1)}| = 2,000$  steps and maximum 6,500 steps across the 8 cells; the single-harmonic detector is therefore the *leading edge* of the multi-harmonic onset, not a narrower or biased estimate. The  $G<F$  ordering label is preserved on all 8 cells under  $m \in \{1, 2, 3, 5\}$ , and the  $\Delta\tau$  gap typically widens (e.g.  $\text{wd}=1.52$ ,  $\text{seed}=42$  shifts from  $\Delta\tau=9,000$  at  $m=1$  to 15,500 at  $m=5$ ). A separate  $K=3$  irrep-mean scoring (an alternative non-degenerate aggregation) gives the same qualitative result on the same 8 cells (`r1_irrep_alignment.csv`). On the logit side, multi-harmonic recompute at the causal-probe checkpoints (Figure 12) shows that the top-1 harmonic captures only 10–30% of the logit Fourier energy while the top-5 captures 60–85% in the post- $\tau_{\text{circuit}}$  regime; this confirms that the full grokked circuit is multi-harmonic but the single-harmonic onset still detects the rise correctly. Extending the same no-break / 1-break / 2-break BIC comparison to all 60 Stage 4  $F_{\text{corr}}$  trajectories plus the 8 post-hoc cells ( $n=68$ ; `results/posthoc/r2_multibreak_extended.csv`), 65/68 cells prefer the 2-break model, but its early break  $\text{bp}_1$  stays within 1,500 steps (3 measurement bins) of the canonical 1-break  $\tau_F$  for 30/68 cells and within 2,500 steps for 46/68; the rare large-displacement cells ( $\geq 5,500$  steps; 7/68) are non-grokking trajectories without a sharply localised early break, where neither the 1-break nor the 2-break fit has a well-posed answer. Across the 8 headline  $G<F$  cells, every 2-break fit’s early breakpoint agrees with the 1-break  $\tau_F$  to within 4,000 steps and never flips the ordering label. A future PELT-based analysis on richer trajectories would localise the saturation knee directly.  **$\tau_{\text{gen}}$  threshold sweep.** At thresholds  $\theta \in \{0.80, 0.85, 0.90, 0.95\}$  on the same 8 cells (sustained over 3 consecutive 500-step log points), the median displacement  $|\tau_{\text{gen}}(\theta) - \tau_{\text{gen}}(0.90)|$  is 1,250 ( $\theta=0.80$ ), 750 ( $\theta=0.85$ ), and 750 ( $\theta=0.95$ ) steps; the maximum is 9,500 steps at  $\theta=0.95$  for one cell whose test accuracy plateau crawls slowly from 0.90 to 0.95. The typical (median) shift is well below the median  $\Delta\tau=6,000$  steps of cells with  $|\Delta\tau| \geq 3,000$  (Section 6.6), so the  $G<F$  ordering at the level of headline 95% CIs is insensitive to the threshold choice across  $[0.80, 0.95]$ . The single 9,500-step worst case occurs

because the run’s test accuracy plateau crawls slowly between 0.90 and 0.95, which is precisely the regime where any  $\tau_{\text{gen}}$  definition becomes fragile; near-coincident cells ( $|\Delta\tau| \leq 2,000$ ) cannot be reliably classified at *any* threshold and are already excluded by the measurement-resolution caveat in Section 6.7. Full per-cell tables are in `results/posthoc/w1_multibreak.csv` and `w5_taugen_threshold.csv`.

**E7 — Causal Fourier-subspace intervention.** Timing alone cannot say whether the subspace our detector tracks does any work. To test whether the dominant low-order Fourier subspace identified by the proxy is causally load-bearing rather than a correlated readout, we probe twelve cells: five original cells spanning the phase diagram (strong  $C < G < F$ ,  $G < C < F$  boundary,  $C < F < G$  slow-grokking, canonical, coincident) plus seven expansion cells varying seed ( $\times 2$ ), weight decay ( $\times 2$ ), learning rate ( $\times 1$ ), task (multiplication,  $\times 1$ ), and prime ( $p=97$ ,  $\times 1$ ). Each cell is saved at three to four regime-boundary checkpoints (pre-all, between events, post-all). At each checkpoint we apply four interventions on the trained logits: **(I1)** keep only the top-3 Fourier harmonics of the output-class index, **(I2)** remove them, **(I3)** remove a matched-energy random subspace as control, and **(I4)** apply the same Fourier removal to the token embedding rows (anonymised supplementary code). All four are representation-space projections evaluated at frozen checkpoints: I1–I3 act on the trained logit matrix with a basis indexed by the answer class, and I4 acts on the embedding rows with a basis indexed by the token value. They answer whether the projected subspace is necessary for test accuracy at a given step; the complementary question of *where* in the network that structure is computed is answered by the label-free internal patch reported at the end of this subsection, whose random control is dimension-matched in the same way as I3.

Four findings emerge (Figure 12 shows a five-cell subset). First, on the eight addition-mod-53 cells that fully grok, the maximum I2 drop across between-event checkpoints reaches 0.40–0.72 in absolute test accuracy (specificity 12–214 $\times$  over matched-energy random controls), demonstrating that the top-3 Fourier subspace is causally load-bearing. The effect extends to  $p=97$ : top-3 alone is insufficient (I2 drop 0.004) because the larger prime spreads load across more harmonics, but at  $K=5$  the drop reaches 0.38 and at  $K=10$  it reaches 1.00, all with random-control drops of exactly 0.000. The multiplication cell (`mul_GF`) gives a clean negative that is informative in its own right: specificity  $\approx 1\times$  at every  $K$ , and the internal patch below reproduces the same null at every site. Multiplication is reindexed through a discrete logarithm, so the cyclic-group harmonics that carry load for addition are not the directions the network uses, even though its *timing* behaviour is indistinguishable from addition (30/30  $G < F$  in Stage 5A). This separates the two scopes cleanly: the embedding-ring lag is a timing regularity that transfers across the tasks we tested, while the identification of *which* subspace bears load is task-specific and holds where the cyclic-group basis is the operative one (addition mod 53 and mod 97). Headline intervention statistics are quoted over the addition cells accordingly. The remaining cell that does not fully grok (`FG_slow`, max test accuracy 0.51) shows correspondingly smaller but still positive I2 effects (0.02–0.39); we include it in Figure 12 for completeness. Second, after  $\tau_F$  the I2 drop on logits collapses while I4 on embeddings stays large, a pattern consistent with redistribution of the answer across the logit representation after  $\tau_F$  while the embedding remains Fourier-dependent. Third, the  $G < C < F$  cell (`G_CF_boundary`,  $\tau_{\text{gen}}=6,500 < \tau_{\text{circuit}}=8,000$ ) shows Fourier load-bearing already at step 7,000 (drop 0.69, specificity 112 $\times$ ), reframing the four  $G < C < F$  exceptions in Section 5.4 as detector-resolution artefacts. Fourth, I4 is not inert before  $\tau_F$ : at checkpoints between  $\tau_{\text{gen}}$  and  $\tau_F$  on the same eight fully-grokking addition cells, removing the top-3 Fourier directions from the token embedding reduces test accuracy from a baseline of 0.90–1.00 to 0.02–0.37 (`results/causal_probe_combined/intervention_results.csv`). Embedding Fourier *content* is therefore functionally necessary well before the BIC-detected consolidation at  $\tau_F$ . This sharpens rather than contradicts the dissociation: what lags generalization is the consolidated global geometry that  $F_{\text{corr}}$  (and a PCA ring plot) detects, not the first functional presence of Fourier structure in the embedding.  $\tau_F$  should accordingly be read throughout as a *consolidation changepoint* rather than a first-appearance time, and the headline claim concerns the late consolidation of the visible ring, not causal irrelevance of the embedding before  $\tau_F$ .

**Why  $K=3$ ? A top- $K$  ablation spectrum.** A reasonable concern is whether  $K=3$  is privileged or whether removing any low-rank subspace would damage accuracy similarly. We rerun the intervention for  $K \in \{1, 3, 5, 10\}$  on all twelve cells, using the same matched-energy random-subspace control at each  $K$ . Table 16 reports medians over the fifteen between-events checkpoints with base accuracy  $> 0.5$  (excluding

`mul_GF`). The causal effect grows smoothly with  $K$  from  $K=1$  (insufficient coverage; I2 drop 0.080) through  $K=3$  (drop 0.551) to  $K=5$  (drop 0.812), and saturates by  $K=10$  (drop 0.974). At every  $K$  the matched-energy random control remains within  $|\Delta \text{acc}| \leq 0.007$  of unintervened accuracy, so the specificity of top- $K$  Fourier removal over a random subspace *rises* from  $10\times$  ( $K=1$ ) to  $\geq 360\times$  ( $K=5$ ). The shape of the curve confirms that the causal effect is localized to low-order logit-Fourier structure rather than to arbitrary low-rank damage.

Table 16: Top- $K$  ablation spectrum on the twelve causal-probe cells (excluding `mul_GF`), restricted to between-events checkpoints with base accuracy  $> 0.5$  ( $n=15$ ). I2 removes the top- $K$  Fourier harmonics; I3 removes a matched-energy random subspace. Specificity is I2 drop / I3 drop.

| $K$ | median I2 drop | median I3 ctrl drop | median specificity       |
|-----|----------------|---------------------|--------------------------|
| 1   | +0.080         | +0.007              | $10.4\times$             |
| 3   | +0.551         | +0.005              | $37.5\times$             |
| 5   | +0.812         | +0.001              | $362\times$              |
| 10  | +0.974         | -0.004              | (saturated) <sup>†</sup> |

<sup>†</sup>At  $K=10$  the random control drop is slightly negative on the median, making the specificity ratio undefined; I2 drop has saturated near 1.0 regardless.

**Are the top- $K$  harmonics special, or would any  $K$  Fourier directions do? A frequency-selectivity control.** The top- $K$  subspace removal still leaves open a second attack: maybe the load-bearing structure is just “any  $K$  Fourier directions”, not specifically the high-power ones. We test this directly by comparing, at the same checkpoints and at  $K=3$ , three alternative ablations that stay *inside* the Fourier basis but pick different frequencies: **top-3** (the three harmonics with the highest centered-logit power, the canonical I2), **alt-3** (three harmonics drawn uniformly from rank  $\geq 10$ , averaged over 5 random draws), and **bot-3** (the three lowest-power harmonics). For comparison we also report the matched-energy non-Fourier random-subspace control (I3) used throughout this section. Table 17 summarises the result over the fifteen between-events checkpoints (excluding `mul_GF`). Top-3 removal drops accuracy by a median of 0.55; alt-3 and bot-3 removals drop accuracy by  $-0.003$  and  $-0.002$  on the median, statistically indistinguishable from the matched-energy non-Fourier random control ( $+0.008$ ). The load-bearing logit-Fourier structure is therefore not “any  $K$  Fourier directions” but specifically the small set of high-power answer-class harmonics.

Table 17: Frequency-selectivity control at  $K=3$  on twelve causal-probe cells (excluding `mul_GF`), restricted to between-events checkpoints with base accuracy  $> 0.5$  ( $n=15$ ). Every variant removes a 3-harmonic Fourier subspace from the centered logits; they differ only in *which* three harmonics are chosen.

| Removed subspace ( $K=3$ )                              | median accuracy drop |
|---------------------------------------------------------|----------------------|
| top-3 Fourier (by power)                                | +0.551               |
| alt-3 Fourier (rank $\geq 10$ , mean of 5 random draws) | -0.003               |
| bot-3 Fourier (lowest-power)                            | -0.002               |
| matched-energy non-Fourier random subspace (I3)         | +0.008               |

**Internal activation patching (F4 pilot).** To address the offline output-space scope of I1–I4, we also ran a *label-free, component-level* pilot on the same 43 probe checkpoints (script `src/analysis/activation_patching_probe.py`, results `results/activation_patching/`). At six internal sites (embedding output, per-layer attention and MLP outputs, final residual stream) we view each activation channel as a function on the  $(a, b)$  input grid, ablate the top-3 grid-Fourier harmonics selected by activation power (no ground-truth label enters the construction), and compare against a dimension-matched random-subspace control mirroring I3 (5 draws per site). On the ten between-event ( $\tau_{\text{gen}} < t < \tau_F$ ) addition-mod-53 checkpoints with base accuracy above 0.5, ablation at the final residual stream drops test accuracy by a median of 0.49 while the random control drops it by 0.02; the second-layer MLP output (`mlp_L1`) shows drops of 0.66 vs 0.10, and second-layer attention (`attn_L1`) 0.77 vs 0.21. On the  $p=97$  cell the separation is sharper still (residual stream 0.44 vs 0.00), and the multiplication cell replicates the E7 negative (ablation  $\approx$  control at

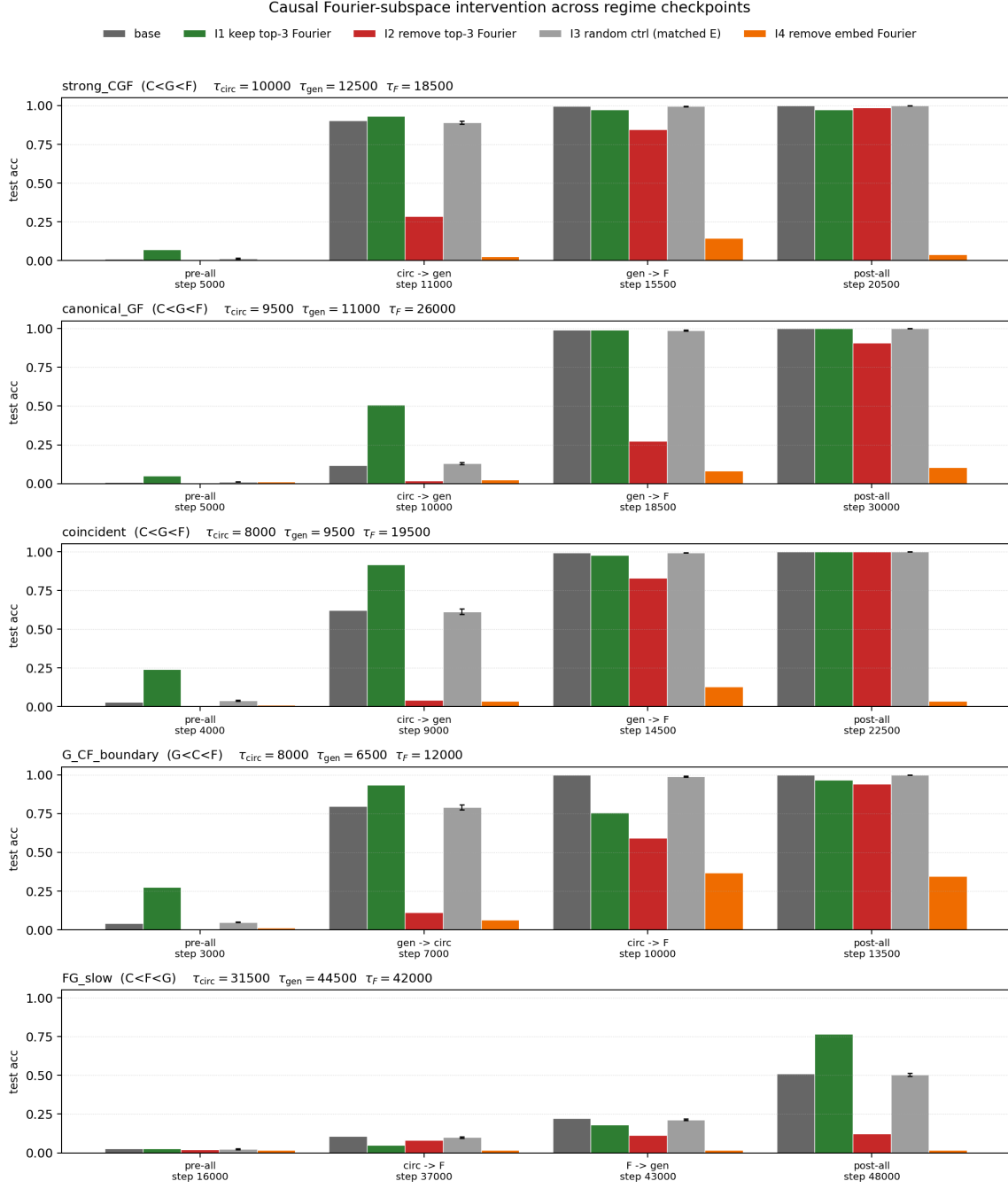


Figure 12: **Causal Fourier-subspace intervention (five-cell subset of the twelve probed cells).** Each row is one cell; the four bar groups are regime checkpoints (*pre-all*, between-event, *post-all*); each group reports test accuracy under five conditions: unintervened **base**; I1 keep only the top-3 Fourier harmonics; I2 remove them; I3 a matched-energy random-subspace control; I4 remove top-3 Fourier from the token embedding. The large I2 drops with negligible I3 effect in the between-event regime (specificity 12–214 $\times$  across all eight fully-grokked addition-mod-53 cells) establish the dominant answer-aligned Fourier subspace as functionally necessary. Note that I4 (orange) is already large at between-event checkpoints: embedding Fourier content is functionally necessary before  $\tau_F$  (see text). The post- $\tau_F$  I2 collapse alongside the sustained I4 drop is consistent with redistribution or redundancy in the logit representation after  $\tau_F$ , while the embedding remains Fourier-dependent.



every site). The grid-space control is the informative comparison at the four sites whose activations mix both input tokens. At the embedding output and the first-layer MLP output (`mlp_L0`), activations are (partially) separable functions of a single token, so a random grid-space direction moves them off the low-dimensional manifold the network ever produces while the axis-aligned Fourier ablation stays on it; the embedding side is accordingly read from I4, whose basis respects that structure. The pilot localizes Fourier-specific functional necessity to the second-layer components and the final residual stream during the  $\tau_{\text{gen}} - \tau_F$  window, adding an internal, label-free probe to the output-space interventions (Appendix E, F4).

The eight post-hoc analyses above, together with the detector sensitivity in Section 6.5, show that the three-stage ordering survives changes to the null-correction parameters, multi-harmonic Fourier variants, idle-step interpolation choices, embedding-geometry definitions, bootstrap sampling uncertainty, the choice of progress measure used to locate  $\tau_{\text{circuit}}$ , the canonical 1-break formulation of  $\tau_F$  versus the post-hoc 2-break sanity check, and the accuracy threshold used to define  $\tau_{\text{gen}}$ . All source scripts and expected artefacts are listed in Appendix A (“Release plan”).

## 6.7 Limitations

**Task scope.** Our three-stage confirmation covers  $(a + b) \bmod p$  and  $(a \cdot b) \bmod 53$ . Stage 5B extends to  $p=97$ . Generalization to non-modular tasks or non-algorithmic settings remains open.

**Architecture scope.** All main experiments use the same 2-layer transformer. A small post-hoc depth pilot at the canonical operating point shows that the ordering does not transfer automatically:  $n_{\text{layers}}=1$  gives  $F < G$  in 3/3 seeds, and  $n_{\text{layers}}=3$  gives  $F < G$  in 2/3 seeds (Appendix, F2). One plausible account is that deeper networks provide alternative pathways that let logit-Fourier structure propagate into embeddings before generalization, collapsing the timing gap. Because this pilot covers only one operating point and two depths, we treat it as a scope warning; a systematic depth  $\times$  width transfer study remains the primary future work.

**Measurement resolution.**  $\Delta\tau$  is measured with 500-step resolution; small lags ( $|\Delta\tau| \lesssim 500$ ) cannot be reliably classified. One  $F < G$  run has  $|\Delta\tau| = 1,500$  (three log-intervals), above the resolution floor but warranting caution.

**Intervention controls.** The logit-side interventions (I2) and the internal activation patch are each paired with a dimension-matched random-subspace control, and both report specificity against that control. The embedding-side intervention I4 is reported without such a pairing: at the embedding and first-layer sites the activations are (partially) separable functions of a single input token, and a random direction drawn in grid-function space moves them off that low-dimensional manifold, so a random control there measures manifold departure rather than Fourier specificity. I4 should accordingly be read as establishing that embedding Fourier content is functionally in use, without a specificity ratio attached; designing on-manifold controls for these sites is part of the systematic study in Appendix E, F4.

**Logit metric alignment.** Our  $F_L$  captures the aggregate Fourier structure of logits across all  $(a, b)$  pairs, which is related but not identical to Nanda et al.’s restricted logit loss. A direct comparison on the same 60 Stage 2+3 runs (E3, Section 6.6, Table 15) shows moderate rank agreement (Spearman  $\rho=0.42\text{--}0.54$ ), confirming that the three detectors track related but not interchangeable aspects of circuit formation.

**Phase-boundary regime.** The 4  $G < C < F$  exceptions in Stage 4 all occur at high lr where generalization is very fast and the BIC changepoint may underestimate  $\tau_{\text{circuit}}$  when the logit signal rises gradually rather than sharply. The causal Fourier-subspace intervention E7 (Section 6.6, Figure 12) supports this detector-lag interpretation on one representative high-lr  $G < C < F$  cell, where the dominant Fourier subspace is already causally load-bearing at step 7,000 even though the BIC-detected  $\tau_{\text{circuit}}$  lands at step 8,000; we have not, however, intervened on all four exceptions, so we treat the detector-lag explanation as supported but not exhaustively verified.



**Cyclic-group structural assumption.** The  $F_{\text{corr}}$  detector projects the embedding matrix onto the 2-dimensional Fourier subspaces of the cyclic group  $\mathbb{Z}/p$  and the  $a \cdot b \bmod p$  extension uses a discrete-log reindexing to the same cyclic-group target. Algorithmic variants that solve modular arithmetic via non-cyclic representations (for example, factor-group decompositions on composite moduli, or learnt non-Fourier features) would not register on  $F_{\text{corr}}$  in its present form, and the  $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$  claim should be read as conditional on the cyclic-group structural prior.

**Residual structure and bootstrap independence.** The BIC changepoint estimator assumes i.i.d. Gaussian residuals on the log-time axis and the bootstrap CI in Table 13 resamples per-run rows as i.i.d. units. The reported intervals should therefore be read as descriptive row-level bootstrap intervals rather than fully valid confidence intervals under clustered dependence across hyperparameter cells: seed-level dependence within a cell is not accounted for by independent-row resampling and may either inflate or deflate the apparent variance. We quantify this gap with a per-source cell-level cluster bootstrap that resamples (lr, wd) cells with replacement and keeps all seeds inside a cell together (Table 14; `src/analysis/cluster_bootstrap.py`). On the three sources where seed clustering is non-degenerate (Stage 3, Stage 6, Stage 4 retraining), the cluster-bootstrap 95% CI for  $P(G < F)$  is 33–40% wider than the row-bootstrap CI; the headline point estimates shift by  $< 1\%$ . We position this comparison as a descriptive bootstrap-uncertainty diagnostic rather than a cluster-robust inferential replacement; Table 13 reports the cluster-bootstrap interval alongside the row-level interval for each source, and the Wilson intervals in Table 5 remain run-level and carry the same caveat. The row-bootstrap intervals are mildly anti-conservative but not qualitatively misleading; where seed clustering matters, the cluster-bootstrap interval is  $\sim 33\text{--}40\%$  wider. Stage 5A, Stage 5B and Stage 2 are at  $P(G < F)$  rates of 100%, 100% and 96.7% respectively, so intra-cluster variance is mechanically near zero and the two bootstrap variants coincide. Heteroscedasticity in  $F_{\text{corr}}$  residuals is not formally tested.

**Detector-specification scope.** The robustness sweeps in Section 6.5 and Section 6.6 vary the detector along axes we identified ahead of time (EMA  $\alpha$ , BIC slope threshold, null refresh interval,  $\tau_{\text{gen}}$  threshold, canonical 1-break formulation versus post-hoc 2-break sanity check). Adversarial detector specifications outside this grid are not ruled out, and an alternative onset definition that rejects single-harmonic projection altogether (for example, group-irrep based detectors) could in principle reorder  $\tau_{\text{circuit}}$  and  $\tau_F$ .

## 7 Ruling Out a Speed-Dependent Explanation of the Ordering

The single strongest alternative account of our result is that  $G < F$  is not a property of how representations form at all, but an artefact of *how fast* the network generalises: if embedding consolidation runs on its own clock, then fast grokking would simply beat that clock and slow grokking would lose to it. This section states that account precisely and tests it directly, using a data-subsampling speed control (E5, Section 5.9), a targeted sweep engineered to enter the regime where the account predicts a reversal, and a leave-one-out predictive diagnostic on the Stage 6 grid. Inside the regime it nominates as decisive, the account’s prediction does not hold.

**The account, stated precisely.** Let  $\tau_{\text{gen}}$  and  $\tau_F$  be determined by two largely separable processes: generalization by the dynamics of the circuit that solves the task, and embedding-geometry consolidation by the indirect reshaping of the embedding rows through gradients back-propagated from the regularised decoder (recall that in our setting  $\text{wd}_{\text{emb}}=0$ , so the embedding has no direct  $L_2$  pressure and is shaped only via the decoder’s weight-decay clock). If these two processes run on roughly parallel clocks, the observed ordering becomes a function of generalization speed:

- when  $\tau_{\text{gen}}$  is small (“fast grokking”), generalization fires before the embedding geometry has had time to consolidate, yielding  $G < F$ ;
- when  $\tau_{\text{gen}}$  is large (“slow grokking”), weight decay has enough time to complete embedding consolidation first, yielding  $F < G$ .

This picture is compatible with Doshi et al. (2024), who show that on corrupted modular-arithmetic data the relative onsets of memorization and generalization depend separately on data corruption, weight decay, and

learning rate; and with the implicit-bias view in which weight decay continues to restructure representations long after the training loss has saturated (Soudry et al., 2018; Galanti et al., 2022; D’Angelo et al., 2024).

**Evidence from existing data.** Three observations in our dataset are consistent with this hypothesis.

(i) *Location of  $F < G$  exceptions.* Both  $F < G$  runs in the 1D sweeps occur at the margins of the Grokking phase where  $\tau_{\text{gen}}$  is unusually late:  $\tau_{\text{gen}}=43,500$  steps ( $\text{lr}=6.8 \times 10^{-4}$ , seed 7;  $\Delta\tau = -1,500$ ) and  $\tau_{\text{gen}}=67,500$  steps ( $\text{lr}=4.32 \times 10^{-3}$ , seed 42;  $\Delta\tau = -54,500$ , max test accuracy 0.965). In Stage 6, 11 of 13  $F < G$  cases lie on the outermost lr rows of the 2D grid, consistent with the same pattern.

(ii) *Cross-task pilot.* A pilot run of modular *subtraction*  $(a-b) \bmod 53$  at the same hyperparameters as our Stage 2 core cell ( $\text{wd}=2.18$ ,  $\text{lr}=1.6 \times 10^{-3}$ ) groks much later than addition ( $\tau_{\text{gen}}=43,500$  steps vs  $\approx 11,000$  steps for addition) and lands in the  $F < G$  regime ( $\Delta\tau=-9,000$ ). A second subtraction seed at the same hyperparameters groks faster ( $\tau_{\text{gen}}=27,500$  steps) and returns to  $G < F$  ( $\Delta\tau=+5,500$ ), providing *within-task, within-hyperparameter* evidence that the ordering flip tracks generalization speed rather than task identity.

(iii) *Aggregate correlation.* Across 57 Grokking runs (55 addition + 2 subtraction pilots),  $\Delta\tau$  and  $\tau_{\text{gen}}$  are negatively correlated ( $r = -0.62$ ; Figure 13). All  $F < G$  runs have  $\tau_{\text{gen}} \geq 43,000$  steps; the majority of  $G < F$  runs have  $\tau_{\text{gen}} \leq 35,000$  steps. Because  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  is defined relative to  $\tau_{\text{gen}}$ , part of this correlation is mechanical and should not be interpreted causally on its own; the within-task seed comparison in (ii) is free of this confound and is what load-bears the hypothesis.

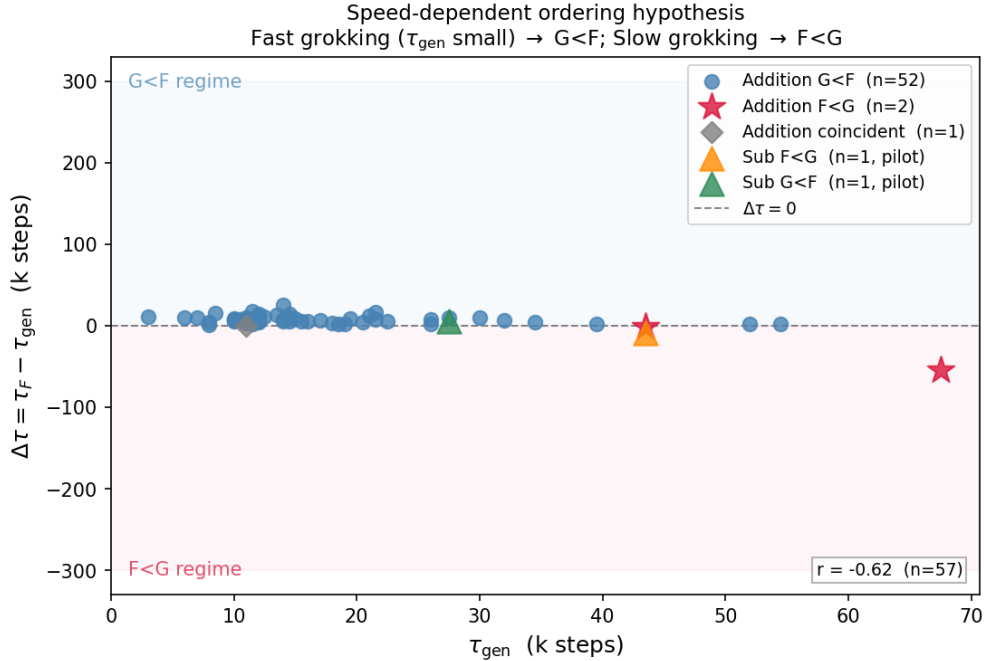


Figure 13: **Speed-dependent ordering.** Scatter of generalization onset  $\tau_{\text{gen}}$  vs lag  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  for all 55 Grokking addition runs (blue circles =  $G < F$ , red stars =  $F < G$ , grey diamond = coincident) and 2 subtraction pilots (orange triangles; 1  $F < G$ , 1  $G < F$ ). All  $F < G$  runs cluster at  $\tau_{\text{gen}} \geq 43,000$  steps; overall  $r = -0.62$  ( $n=57$ ). The two subtraction pilots at identical hyperparameters ( $\text{wd}=2.18$ ,  $\text{lr}=1.6 \times 10^{-3}$ ) differ only in seed and generalization speed, yet flip ordering—within-task evidence that the effect is not a pure  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  construction artefact.

**Leave-one-out predictive diagnostic.** A leave-one-out logistic regression on the 123 Stage 6 cells with both  $\tau_{\text{gen}}$  and  $\tau_F$  detected and  $|\Delta\tau| > 500$ , using  $(\log_{10} \text{lr}, \log_{10} \text{wd}, \log(1 + \tau_{\text{gen}}))$  as features and  $\text{sign}(\Delta\tau)$  as target, achieves ROC-AUC 0.945 (95% bootstrap CI [0.896, 0.981],  $n_{\text{boot}}=1000$ ); restricting the

model to  $(\log_{10} \text{lr}, \log_{10} \text{wd})$  alone drops AUC to 0.549, while  $\log(1 + \tau_{\text{gen}})$  alone retains 0.925. We frame this as a *predictive diagnostic* consistent with the speed-dependent hypothesis, not as an independent causal test: the label  $\text{sign}(\Delta\tau)$  contains  $\tau_{\text{gen}}$  by construction, so even under a null where  $\tau_F$  and  $\tau_{\text{gen}}$  are independent the AUC could be substantial whenever  $\tau_{\text{gen}}$  varies over a range comparable to the support of  $\tau_F$ , which is the regime our Stage 6 grid happens to inhabit. What the regression rules out is the alternative that the  $(\text{lr}, \text{wd})$  coordinates predict the ordering through a route distinct from  $\tau_{\text{gen}}$ : the marginal 0.020 AUC lift from adding  $(\log_{10} \text{lr}, \log_{10} \text{wd})$  on top of  $\log(1 + \tau_{\text{gen}})$  is small, supporting a single-scalar speed description of the ordering. The predictive diagnostic is monotone-consistent with the speed-dependent picture but does not on its own break the mechanical entanglement between  $\tau_{\text{gen}}$  and  $\Delta\tau$ ; the test that does provide independent evidence is the targeted slow-grokking sweep below, which directly falsifies the strong form of the speed-dependent regime. Per-cell predictions are in `results/posthoc/r6_speed_predictive.csv`.

**What would falsify this hypothesis.** A clean test would hold the weight-decay schedule fixed and independently manipulate generalization speed without altering the weight-decay clock, then check whether the ordering label flips once  $\tau_{\text{gen}}$  crosses the slow-grokking margin. The strong form of the hypothesis predicts a monotonic transition from  $G < F$  to  $F < G$  as  $\tau_{\text{gen}}$  exceeds a threshold around 35,000 steps (the two original  $F < G$  outliers had  $\tau_{\text{gen}} \geq 43,500$ ); the data-subsampling control (Section 5.9) does not reach this margin because all  $\tau_{\text{gen}}$  values there sit below 13,000.

**Targeted slow-grokking sweep falsifies the strong form.** We therefore ran a targeted sweep that fixes  $(\text{wd}, \text{seeds})$  to  $(2.5, \{42, 7, 2025\})$  and sweeps low learning rates  $\text{lr} \in \{5.0, 5.5, 6.0, 6.5, 6.8, 7.2, 7.6, 8.0\} \times 10^{-4}$  to 100,000 steps (auto-extending to 120,000 if needed), yielding 24 cells densely covering the slow-grokking margin (`src/sweeps/slow_grokking_sweep.py`, `results/slow_grokking/results.csv`). **All 24 cells grok and all 24 are  $G < F$** , with  $\Delta\tau$  ranging from +2,500 to +19,500 steps (median +11,750); 12 of the 24 cells reach  $\tau_{\text{gen}} \geq 35,000$ , and the slowest ( $\tau_{\text{gen}} = 54,500$ ,  $\text{lr} = 5 \times 10^{-4}$ ,  $\text{seed} = 7$ ) still reports  $\Delta\tau = +3,000$ . A tercile breakdown by  $\tau_{\text{gen}}$  confirms the absence of the predicted transition: 0/9, 0/7, 0/8  $F < G$  in the low, middle, and high terciles respectively (Figure 14). The strong form of the speed-dependent hypothesis is therefore *empirically falsified* on this targeted slice.

A consistent finding emerges for the original Stage 3  $F < G$  outlier at  $(\text{lr}, \text{seed}) = (6.8 \times 10^{-4}, 7)$ : the original Stage 3 sweep reported  $\tau_{\text{gen}} = 43,500$  and  $\Delta\tau = -1,500$  ( $F < G$ ), but the same nominal  $(\text{lr}, \text{wd}, \text{seed})$  re-trained on different hardware in this slow-grokking sweep yields  $\tau_{\text{gen}} = 38,500$  and  $\Delta\tau = +9,000$  ( $G < F$ ), a 10,500-step shift in  $\Delta\tau$  and a flipped ordering label. Combined with the Stage 4 vs E3 cross-hardware comparison (Section 6.6, E3), this strongly suggests the two  $F < G$  ordering outliers in Stage 3 are seed-level / hardware-level fluctuations rather than a systematic regime transition.

**What this establishes.** Across every regime we can reach, generalization speed does not determine the ordering. The  $G < F$  ordering is invariant to  $\tau_{\text{gen}}$  both at the canonical operating point, where a data-subsampling control moves  $\tau_{\text{gen}}$  without moving the ordering (E5, Section 5.9), and in the slow-grokking regime that the speed account identifies as its decisive test, where 24/24 purpose-built cells remain  $G < F$  out to  $\tau_{\text{gen}} = 54,500$  steps. The two original  $F < G$  points that motivated the account are themselves reproduced as  $G < F$  when re-trained, so they carry no weight against it either. These 24 cells extend the aggregate to 236/260=90.8%  $G < F$ . Within the range of generalization speeds our setup can produce, the embedding-ring lag behaves as a property of how the two representations form rather than as a race between two clocks.

## 8 Conclusion

The Fourier ring in a grokked transformer’s embedding PCA is a lagging indicator of the algorithm it is taken to depict. By measuring answer-aligned logit Fourier alignment  $\tau_{\text{circuit}}$ , generalization  $\tau_{\text{gen}}$ , and embedding-Fourier consolidation  $\tau_F$  as changepoints within the same runs, we recover the ordering  $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$  in 48/54 runs, with roughly equal spacing between consecutive events ( $\approx 6,000$ – $6,250$  steps each) and a median embedding-geometry lag that scales with task difficulty, from  $\approx 6,000$  steps at  $p=53$  to 28,000 at  $p=97$ .

The ordering is functional, not cosmetic. Removing the dominant answer-aligned Fourier subspace between  $\tau_{\text{circuit}}$  and  $\tau_F$  costs 0.40–0.72 test accuracy while matched-energy random removals cost nothing measurable,

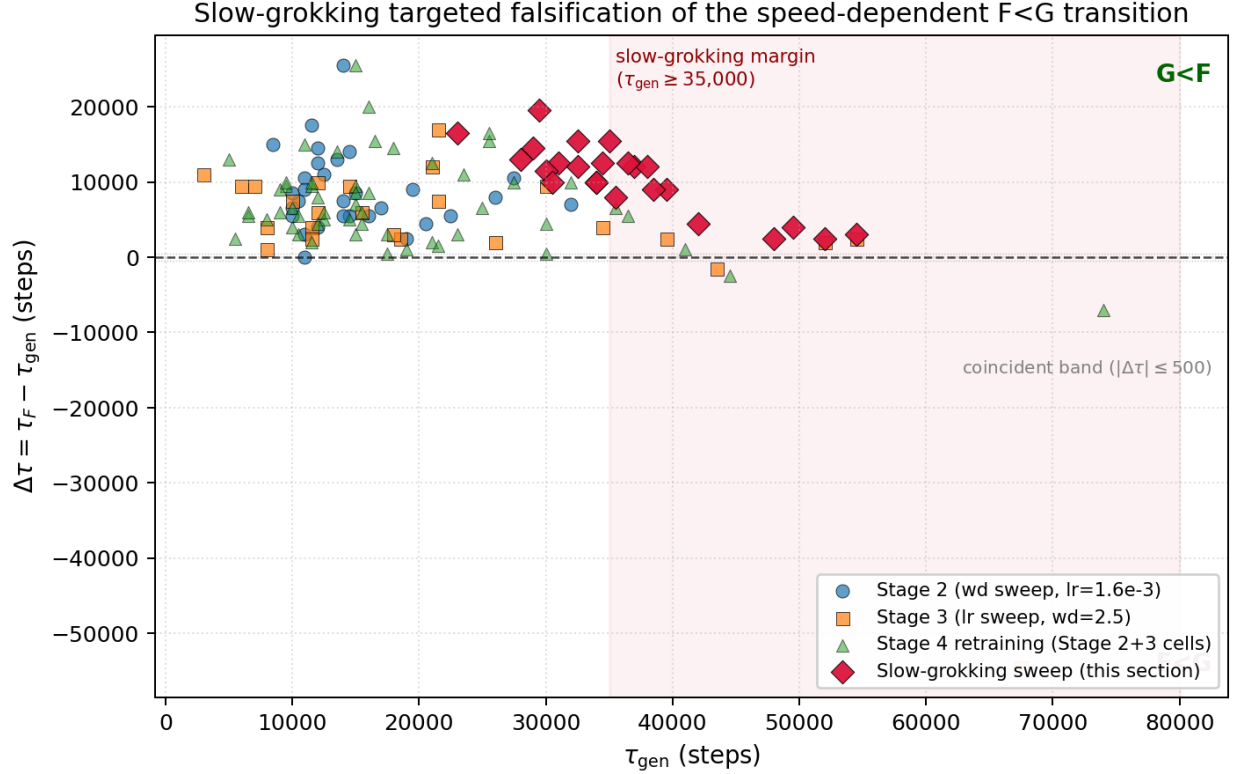


Figure 14: **Targeted slow-grokking sweep falsifies the strong form.** Scatter of  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  versus  $\tau_{\text{gen}}$  for all Grokking cells across four sources. The crimson diamonds are the targeted slow-grokking sweep (24 cells,  $\text{lr} \in [5.0, 8.0] \times 10^{-4}$ ,  $\text{wd} = 2.5$ , seeds  $\{42, 7, 2025\}$ , max steps 100,000–120,000); blue circles are Stage 2 (wd sweep,  $\text{lr} = 1.6 \times 10^{-3}$ ); orange squares are the original Stage 3 (lr sweep,  $\text{wd} = 2.5$ ); green triangles are the Stage 4 retraining of the Stage 2+3 cells. The shaded region marks the slow-grokking margin ( $\tau_{\text{gen}} \geq 35,000$ ). All 24 targeted cells fall in the  $G < F$  half plane ( $\Delta\tau > 0$ ); the 12 cells inside the shaded margin are uniformly  $G < F$ . The original Stage 3  $F < G$  points (orange squares with  $\Delta\tau < 0$ ) are not replicated by the cross-hardware re-runs visible elsewhere on the plot, consistent with seed/hardware variance rather than a systematic regime transition.

and a label-free internal patch reproduces that specificity at the second-layer components and the final residual stream. The same removal applied to the token embedding also bites before  $\tau_F$ : what consolidates late is the visible global geometry, not the functional Fourier content.

The lag is a stable feature of this setting rather than an artefact of one slice. It holds in 212/236 Grokking runs across two tasks, two primes, a dense (lr, wd) grid, and every weight-decay and cross-task slice without exception (29/30 with one tie, 30/30, 21/21); a purpose-built slow-grokking sweep closes off generalization speed as the driver, and a 36-configuration sweep of the changepoint detector’s own hyperparameters leaves the ordering unchanged wherever  $|\Delta\tau| \geq 3,000$  steps.

The practical consequence is a caution about a widely used diagnostic: embedding geometry read as a progress measure systematically understates how far the computation has advanced. A clean ring means the algorithm arrived some thousands of steps ago, not that it is arriving now.

### Broader Impact Statement

This work studies a small-scale neural network training phenomenon. No immediate harmful applications are foreseen. Findings may inform training efficiency research by clarifying which structural changes are or are not necessary for generalization.

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## A Experimental Details

### A.1 BIC Changepoint Estimator

We apply an exponential moving average ( $\alpha=0.15$ ) to the  $F_{\text{corr}}$  trajectory, then operate on  $\log(1+t)$ . We search all candidate breakpoints in the interior two-thirds of the sequence, fitting OLS regressions before and after each candidate. The 4-parameter breakpoint model is accepted if  $\text{BIC}_{\text{break}} < \text{BIC}_{\text{no-break}}$ , where  $\text{BIC}_k = n \log(\text{RSS}/n) + k \log(n)$ . A persistence check requires the post-break slope to exceed 1% of the total value range.

### A.2 Null Permutation Estimation

Each run maintains a fixed NumPy RNG seeded as  $\text{run\_seed} \oplus 0\text{xDEAD}$ , independent of the model RNG. We shuffle the  $p$  token indices 100 times at every 500-step log point (the canonical refresh cadence; see Section 4) to estimate  $F_{\text{null},p95}$ . Sensitivity-analysis configurations using 2,500- and 5,000-step refresh intervals are reported in Section 6.5.

### A.3 Hyperparameter Values

**Stage 2.**  $\text{wd} \in \{1.20, 1.35, 1.52, 1.71, 1.93, 2.18, 2.45, 2.76, 3.11, 3.50\}$ ;  $\text{lr} = 1.6 \times 10^{-3}$ ;  $\text{seeds} = \{42, 7, 2025\}$ .

**Stage 3.**  $\text{lr} \in \{5.00, 6.80, 9.26, 12.6, 17.1, 23.3, 31.7, 43.2, 58.8, 80.0\} \times 10^{-4}$ ;  $\text{wd} = 2.5$ ;  $\text{seeds} = \{42, 7, 2025\}$ .

## B Complete Timing Results

Figure 15 shows the full scatter of  $(\tau_{\text{gen}}, \tau_F)$  for all 55 Grokking runs. Points above the diagonal ( $\tau_F > \tau_{\text{gen}}$ ) correspond to  $G < F$ ; the two red star (★) points below are the  $F < G$  cases discussed in Section 5.2. The scatter confirms that the lag  $\Delta\tau = \tau_F - \tau_{\text{gen}}$  is distributed throughout the range 1,000–25,000 steps with no obvious bimodal structure.

## C Per-Cell Detector Sensitivity

Figure 16 extends the aggregate robustness heatmap (Figure 10) to show  $P(G < F)$  per cell, allowing inspection of which configurations drive the  $\text{RS} < 1$  cases.

Figure 17 provides a direct comparison of the  $F < G$  outlier trajectory against the strong  $G < F$  reference cell.

## D Reproducibility

The majority of experiments reported in this paper were run on a local NVIDIA RTX 4080 Laptop (12 GB VRAM); a smaller subset of longer-running sweeps was executed on rented cloud GPU instances (NVIDIA RTX PRO 6000 and NVIDIA A800) accessed remotely via SSH. All runs used the same PyTorch training loop

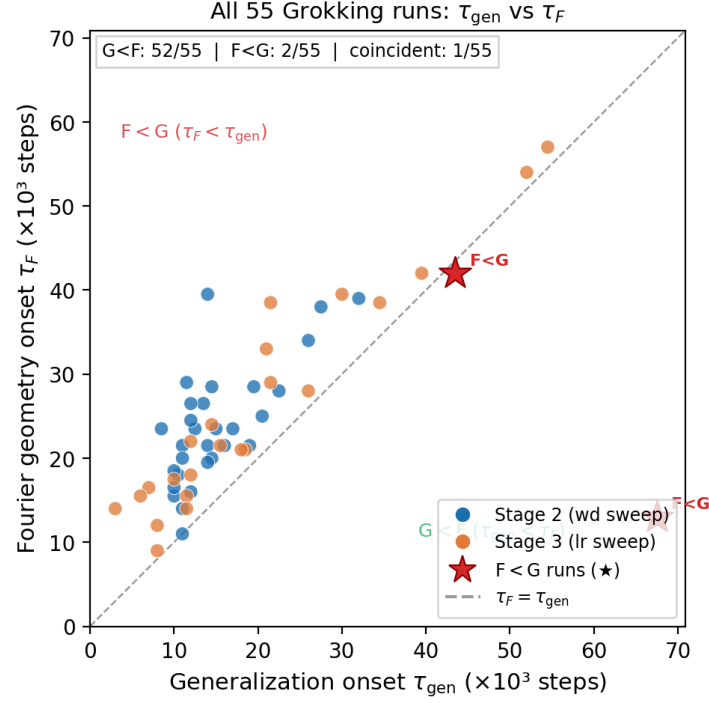


Figure 15: Scatter plot of  $\tau_{\text{gen}}$  vs  $\tau_F$  for all 55 Grokking runs (blue = Stage 2 wd sweep; orange = Stage 3 lr sweep). All points above the dashed diagonal are  $G < F$ . The two red star ( $\star$ ) points below the diagonal are the  $F < G$  outliers discussed in Section 5.2.

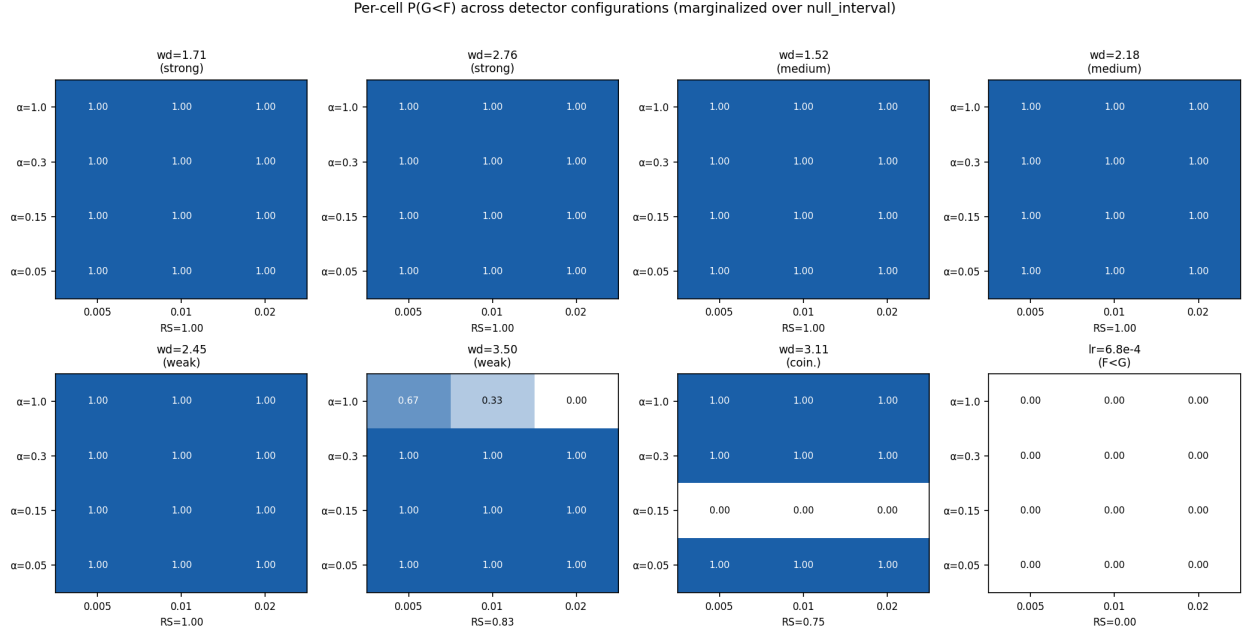


Figure 16: Per-cell  $P(G < F)$  across all 36 detector configurations, marginalized over null-update interval. Strong and medium  $G < F$  cells (first four panels) are uniformly blue ( $RS = 1.00$ ). Weak  $G < F$  cells degrade to *coincident* (not  $F < G$ ) under the strictest slope threshold. The  $F < G$  outlier (bottom right) is uniformly red across all 36 configurations, confirming that its ordering is stable with respect to detector parameters.

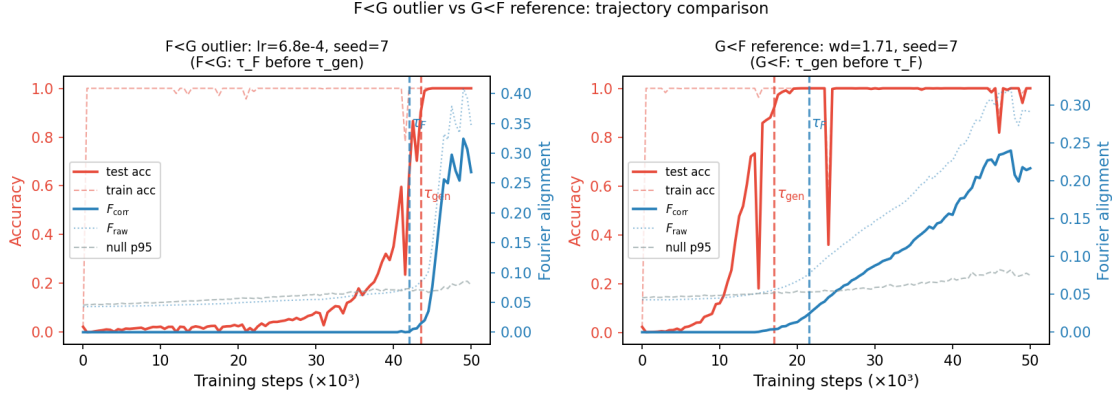


Figure 17: Trajectory comparison: F<G outlier ( $lr=6.8\times 10^{-4}$ , seed 7; left) vs. a strong G<F reference ( $wd=1.71$ , seed 7; right). In the outlier,  $F_{raw}$  (dotted blue) hovers near the null baseline (grey dashed) throughout training and  $F_{corr}$  only barely exceeds zero near the very late  $\tau_{gen} = 43,500$ . In the reference cell,  $F_{corr}$  shows a clear, sustained rise well after the sharp test-accuracy jump.

(Python 3.11.9, PyTorch 2.5.1+cu121, NumPy 2.4.3, CUDA 12.1, cuDNN 9.1.0) with identical model and optimiser code, and `torch.backends.cudnn.deterministic=True` is set in the seeding routine. **Bit-exact reproducibility holds within a single GPU/CUDA configuration but not across them.** A direct comparison of two independent sweeps on the same nominal seeds (Stage 4 on a local RTX 4080 vs E3 on a rented A800), documented in Section 6.6 (E3), shows per-cell  $\tau_{gen}$  shifting by a median of 2,000 steps and the three-way ordering label reclassifying on 8/60 cells. Single-run point estimates of C<G and C<G<F rates therefore carry an empirically observed  $\sim 9$  percentage points of cross-hardware run-to-run variance. Each modular-arithmetic training run terminates in 20–45 minutes at 50,000 steps and 30–75 minutes at 80,000 steps on the local 4080; cloud runs are correspondingly faster on the larger data-centre GPUs. The full set of experiments reported here (Stages 2–6 plus the subtraction pilots) totals approximately 280 runs and  $\sim 170$  GPU-hours (aggregated across hardware).

**Random seeds.** Each cell is run with seeds  $\{42, 7, 2025\}$ . Seeds are applied uniformly to PyTorch, NumPy, and Python’s `random` at the start of each run. The permutation-null estimator uses an independent NumPy generator seeded with `run_seed  $\oplus$  0xDEAD` so that null resampling is reproducible but decoupled from model initialisation.

**Hyperparameter table.** Full hyperparameter values for each stage are listed in Appendix A. The canonical detector configuration used for all headline numbers is: null-update interval = 500 steps, EMA smoothing  $\alpha = 0.15$ , BIC slope threshold = 0.01 of the total value range.

**Release plan.** An anonymized supplementary package for review includes the training scripts, per-run metric trajectories ( $F_{corr}$ ,  $F_L$ , test accuracy at every 500-step checkpoint, stored as compressed `.npz` files), changepoint scripts, plotting scripts that reproduce every figure and table in this paper, `requirements.txt`, and configuration files for every axis of variation reported. A de-anonymized public repository will be released after acceptance.

## E Future Work

We list four natural next steps that extend the empirical scope of this paper. None is required to support the claims in the main body, and all are deferred to future work.

**F1. Full subtraction sweep.** Replicate Stages 2 and 3 with operation  $(a - b) \bmod p$  for  $p \in \{53, 97\}$ . Expected cost:  $\sim 60$  runs,  $\sim 45$  GPU-hours. The G<F prediction is that the rate should track  $\tau_{\text{gen}}$  rather than the task label.

**F2. Architecture-depth scan.** The depth pilot summarised in the Limitations (Section 6.7, “Architecture scope”) shows that the G<F ordering reverses at  $n_{\text{layers}} \in \{1, 3\}$  on the single canonical operating point tested. A full scan should sweep depth and width jointly across multiple Grokking cells, rather than only the single canonical operating point used in `results/arch_depth/results.csv`.

**F3. Online embedding intervention.** The causal interventions in E7 (Section 6.6) operate on static checkpoints: we project out Fourier subspaces at inference time and measure the accuracy drop. A stronger test would intervene *during training*, clamping embedding gradients to the null space of the dominant Fourier harmonics and observing whether generalization is delayed, preserved, or destroyed. This would distinguish whether the embedding ring is a passive by-product of decoder weight decay or plays an active role in stabilising the learned circuit.

**F4. Component-level activation patching.** The E7 interventions act on output logits (with a basis indexed by the ground-truth answer class) and on embedding rows (with a basis indexed by the token value); they establish functional necessity of the projected subspaces but do not localize where inside the network the structure is computed. A label-free, component-level pilot on all 43 probe checkpoints (reported in Section 6.6, E7, “Internal activation patching”) localizes Fourier-specific functional necessity to the second-layer components and the final residual stream during the  $\tau_{\text{gen}} - \tau_F$  window. A systematic version would extend this across the full cell grid and across depths, resolve per-head (rather than per-module) contributions, and design on-manifold controls for the embedding and first-layer sites where a grid-space random control is inherently destructive. The probe script is included in the supplementary code (`src/analysis/activation_patching_probe.py`).